

REDOX REACTION

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THEORY**Oxidation & Reduction**

Let us do a comparative study of oxidation and reduction :

Oxidation	Reduction
1. Addition of Oxygen e.g. $2\text{Mg} + \text{O}_2 \rightarrow 2\text{MgO}$	1. Removal of Oxygen e.g. $\text{CuO} + \text{C} \rightarrow \text{Cu} + \text{CO}$
2. Removal of Hydrogen e.g. $\text{H}_2\text{S} + \text{Cl}_2 \rightarrow 2\text{HCl} + \text{S}$	2. Addition of Hydrogen e.g. $\text{S} + \text{H}_2 \rightarrow \text{H}_2\text{S}$
3. Increase in positive charge e.g. $\text{Fe}^{2+} \rightarrow \text{Fe}^{3+} + \text{e}^-$	3. Decrease in positive charge e.g. $\text{Fe}^{3+} + \text{e}^- \rightarrow \text{Fe}^{2+}$
4. Increase in oxidation number (+2) (+4) e.g. $\text{SnCl}_2 \rightarrow \text{SnCl}_4$	4. Decrease in oxidation number (+7) (+2) e.g. $\text{MnO}_4^- \rightarrow \text{Mn}^{2+}$
5. Removal of electron e.g. $\text{Sn}^{2+} \rightarrow \text{Sn}^{4+} + 2\text{e}^-$	5. Addition of electron e.g. $\text{Fe}^{3+} + \text{e}^- \rightarrow \text{Fe}^{2+}$

Oxidation Number

- It is an imaginary or apparent charge developed over atom of an element when it goes from its elemental free state to combined state in molecules.
- It is calculated on basis of an arbitrary set of rules.
- It is a relative charge in a particular bonded state.
- In order to keep track of electron-shifts in chemical reactions involving formation of compounds, a more practical method of using oxidation number has been developed.
- In this method, it is always assumed that there is a complete transfer of electron from a less electronegative atom to a more electronegative atom.

Rules governing oxidation number

The following rules are helpful in calculating oxidation number of the elements in their different compounds. it is to be remembered that the basis of these rule is the electronegativity of the element.

○ Fluorine atom :

Fluorine is most electronegative atom (known). It always has oxidation number equal to -1 in all its compounds

○ Oxygen atom :

In general and as well as in its oxides, oxygen atom has oxidation number equal to -2 .

In case of

(i) peroxide (e.g. H_2O_2 , Na_2O_2) is -1 ,

(ii) super oxide (e.g. KO_2) is $-1/2$

(iii) ozonide (e.g. KO_3) is $-1/3$

(iv) in OF_2 is $+2$ & in O_2F_2 is $+1$

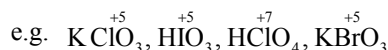
○ Hydrogen atom :

In general, H atom has oxidation number equal to $+1$. But in metallic hydrides (e.g. NaH , KH), it is -1 .

○ Halogen atom :

In general, all halogen atoms (Cl, Br, I) have oxidation number equal to -1 .

But if halogen atom is attached with a more electronegative atom than halogen atom, then it will show positive oxidation numbers.

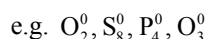


Metals :

- (a) Alkali metal (Li, Na, K, Rb,) always have oxidation number +1
- (b) Alkaline earth metal (Be, Mg, Ca) always have oxidation number +2.
- (c) Aluminium always has +3 oxidation number

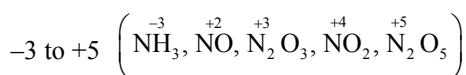
Note : Metal may have negative or zero oxidation number

- Oxidation number of an element in free state or in allotropic forms is always zero



- Sum of the oxidation numbers of atoms of all elements in a molecule is zero.
- Sum of the oxidation numbers of atoms of all elements in an ion is equal to the charge on the ion.
- If the group number of an element in modern periodic table is n, then its oxidation number may vary from (n – 10) to (n – 18) (but it is mainly applicable for p-block elements)

e.g. N-atom belongs to 15th group in the periodic table, therefore as per rule, its oxidation number may vary from



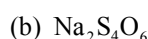
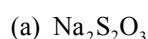
- The maximum possible oxidation number of any element in a compound is never more than the number of electrons in valence shell. (but it is mainly applicable for p-block elements)

Calculation of average oxidation number :

Solved Examples

Example-1 :

Calculate oxidation number of underlined element :



Sol. (a) Let oxidation number of S-atom is x. Now work accordingly with the rules given before.

$$(+1) \times 2 + (x) \times 2 + (-2) \times 3 = 0$$

$$x = +2$$

(b) Let oxidation number of S-atom is x

$$\therefore (+1) \times 2 + (x) \times 4 + (-2) \times 6 = 0$$

$$x = +2.5$$

- It is important to note here that $\text{Na}_2\underline{\text{S}}_2\text{O}_3$ have two S-atoms and there are four S-atom in $\text{Na}_2\underline{\text{S}}_4\text{O}_6$. However none of the sulphur atoms in both the compounds have +2 or +2.5 oxidation number, it is the average of oxidation number, which reside on each sulphur atom. Therefore, we should work to calculate the individual oxidation number of each sulphur atom in these compounds.

Calculation of individual oxidation number

It is important to note that to calculate individual oxidation number of the element in its compound one should know the structure of the compound and use the following guidelines.

Formula :

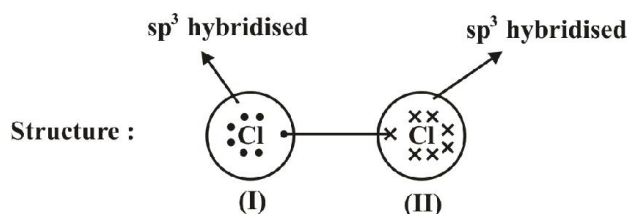
Oxidation Number = Number of electrons in the valence shell - Number of electrons taken up after bonding

Guidelines : It is based on electronegativity of elements.

1. If there is a bond between similar type of atom and each atom has same type of hybridisation, then bonded pair electrons are equally shared by each element.

Example :

Calculate oxidation number of each Cl-atom in Cl_2 molecule



I : Number of electrons in the valence shell = 7

Number of electrons taken up after bonding = 7.

$\therefore$ oxidation number = $7 - 7 = 0$.

II : similarly, oxidation number = $7 - 7 = 0$

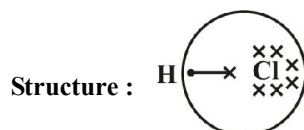
2. If there is a bond between different type of atoms :

e.g. A – B (if B is more electronegative than A)

Then after bonding, bonded pair of electrons are counted with B-atom

Example :

Calculate oxidation number of each atom in HCl molecule



Note : Electron of H-atom is now counted with Cl-atom, because Cl-atom is more electronegative than H-atom

H : Number of electrons in the valence shell = 1

Number of electrons taken up after bonding = 0

Oxidation number of H = $1 - 0 = +1$

Cl : Number of electrons in the valence shell = 7

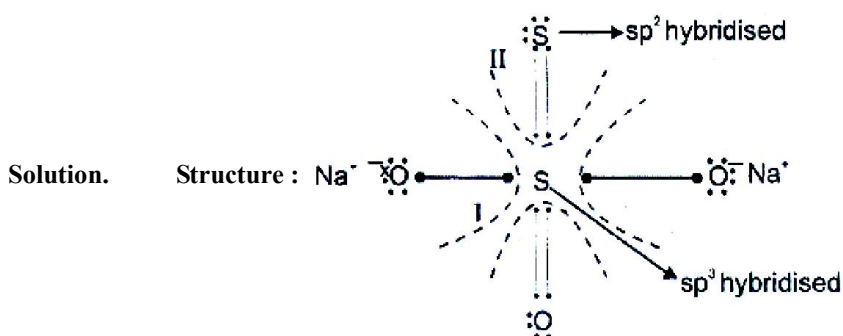
Number of electrons taken up after bonding = 8

Oxidation number of Cl = $7 - 8 = -1$

Solved Examples

Example - 2

Calculate individual oxidation number of each S-atom in $\text{Na}_2\text{S}_2\text{O}_3$ (sodium thiosulphate) with the help of its structure.



Note : I (central S-atom) is sp^3 hybridised (25% s-character) and II (terminal S-atom) is sp^2 hybridised (33% s-character). Therefore, terminal sulphur atom is more electronegative than central sulphur atom. Now, the shared pair of electrons are counted with terminal S-atom.

$\therefore$ I, S-atom : Number of electrons in the valence shell = 6

Number of electrons left after bonding = 0

Oxidation number of central S-atom = $6 - 0 = +6$

II, S-atom : Number of electrons in the valence shell = 6

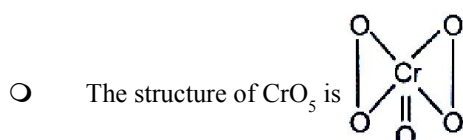
Number of electrons left after bonding = 8

Oxidation number of terminal S-atom = $6 - 8 = -2$

Now, you can also calculate Average Oxidation number of S = $\frac{6 + (-2)}{2} = +2$ (as we have calculated before)

Miscellaneous Examples :

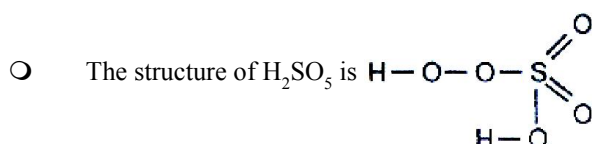
In order to determine the exact or individual oxidation number we need to take help from the structures of the molecules. Some special cases are discussed as follows :



From the structure, it is evident that in CrO_5 there are two peroxide linkages and one double bond. The contribution of each peroxide linkage is -2 . Let the oxidation number of Cr is x .

$\therefore x + (-2) \times 2 + (-2) = 0$ or $x = 6$

$\therefore$ Oxidation number of Cr = $+6$ **Ans.**



From the structure, it is evident that in H_2SO_5 , there is one peroxide linkage, two sulphur-oxygen double bonds and one OH group. Let the oxidation number of S = x .

$$\therefore (+1) + (-2) + x + (-2) 2 + (-2) + 1 = 0$$

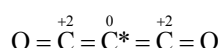
$$\text{or } x + 2 - 8 \quad \text{or } x - 6 = 0 \quad \text{or } x = 6$$

$\therefore$ Oxidation number of S in H_2SO_5 is + 6 Ans.

Paradox of fractional oxidation number

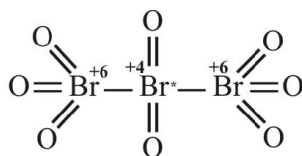
Fractional oxidation number is the average of oxidation state of all atoms of element under examination and the structural parameters reveal that the atoms of element for whom fractional oxidation state is realised are actually present in different oxidation states. Structure of the species C_3O_2 , Br_3O_8 and $\text{S}_4\text{O}_6^{2-}$ reveal the following bonding situations :

- The element marked with asterisk (*) in each species is exhibiting different oxidation number from rest of the atoms of the same element in each of the species. This reveals that in C_3O_2 , two carbon atoms are present in +2 oxidation state each whereas the third one is present in zero oxidation state and the average is +4/3. However, the realistic picture is +2 for two terminal carbons and zero for the middle carbon.



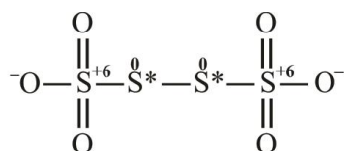
Structure of C_3O_2 (Carbon suboxide)

- Likewise in Br_3O_8 , each of the two terminal bromine atoms are present in +6 oxidation state and the middle bromine* is present in +4 oxidation state. Once again the average, that is different from reality, is + 16/3.



Structure of Br_3O_8 (Tribromooctaoxide)

- In the same fashion, in the species $\text{S}_4\text{O}_6^{2-}$, average oxidation number of S is + 2.5, whereas the reality being + 5, 0*, 0* and +5 oxidation number respectively for respective sulphur atoms.



Structure of $\text{S}_4\text{O}_6^{2-}$, (tetrathionate ion)

In general, the conclusion is that the idea of fractional oxidation state should be taken with care and the reality is revealed by the structures only.

Oxidising and reducing agent

Oxidising agent or Oxidant :

Oxidising agents are those compounds which can oxidise others and reduce itself during the chemical reaction. Those reagents in which for an element, oxidation number decreases or which undergoes gain of electrons in a redox reaction are termed as oxidants.

e.g. KMnO_4 , $\text{K}_2\text{Cr}_2\text{O}_7$, HNO_3 , conc. H_2SO_4 etc are powerful oxidising agents.

REDOX REACTION

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○ Reducing agent or Reductant :

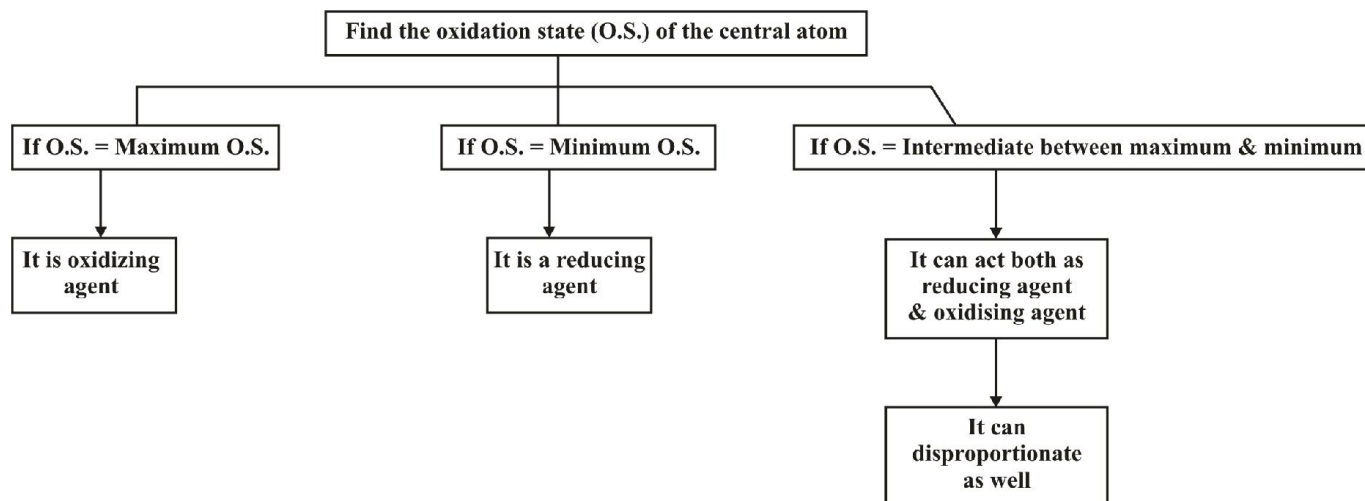
Reducing agents are those compounds which can reduce other and oxidise itself during the chemical reaction. Those reagents in which for an element, oxidation number increases or which undergoes loss of electrons in a redox reaction are termed as reductants.

e.g. KI, $\text{Na}_2\text{S}_2\text{O}_3$ etc are the powerful reducing agents.

Note : There are some compounds also which can work both as oxidising agent and reducing agent

e.g. H_2O_2 , NO_2^-

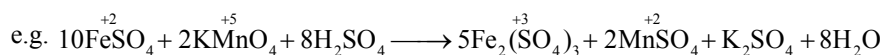
HOW TO IDENTIFY WHETHER A PARTICULAR SUBSTANCE IS AN OXIDISING OR A REDUCING AGENT



Redox reaction

A reaction in which oxidation and reduction simultaneously take place is called a redox reaction

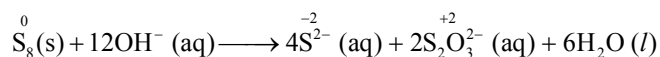
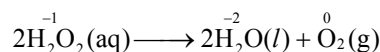
In all redox reactions, the total increase in oxidation number must be equal to the total decrease in oxidation number.



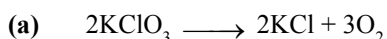
Disproportionation Reaction :

A redox reaction in which same element present in a particular compound in a definite oxidation state is oxidized as well as reduced simultaneously is a disproportionation reaction.

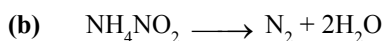
Disproportionation reactions are a special type of redox reactions. One of the reactants in a disproportionation reaction always contains **an element that can exist in at least three oxidation states**. The element in the form of reacting substance is in the intermediate oxidation state and both higher and lower oxidation states of that element are formed in the reaction. For example :



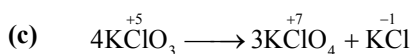
Consider the following reactions :



KClO_3 plays a role of oxidant and reductant both. Here, Cl present in KClO_3 is reduced and O present in KClO_3 is oxidized. Since same element is not oxidized and reduced, so it is not a disproportionation reaction, although it looks like one.



Nitrogen in this compound has -3 and +3 oxidation number, which is not a definite value. So it is not a disproportionation reaction. It is an example of comproportionation reaction, which is a class of redox reaction in which an element from two different oxidation state gets converted into a single oxidation state.

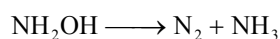
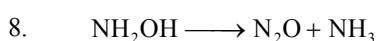
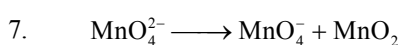
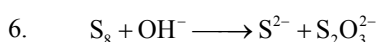
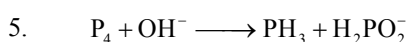
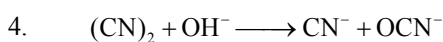
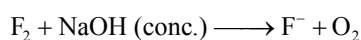
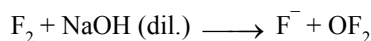


It is a case of disproportionation reaction and Cl atom is disproportionating.

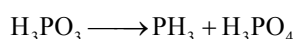
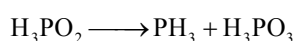
List of some important disproportionation reactions

1. $\text{H}_2\text{O}_2 \longrightarrow \text{H}_2\text{O} + \text{O}_2$
2. $\text{X}_2 + \text{OH}^- (\text{dil.}) \longrightarrow \text{X}^- + \text{XO}^- (\text{X} = \text{Cl, Br, I})$
3. $\text{X}_2 + \text{OH}^- (\text{conc.}) \longrightarrow \text{X}^- + \text{XO}_3^-$

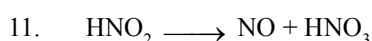
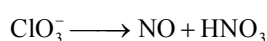
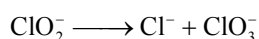
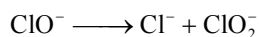
F₂ does not undergo disproportionation as it is the most electronegative element.



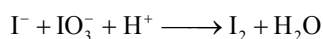
9. Oxyacids of Phosphorus (+1, +3 oxidation number)



10. Oxyacids of Chlorine (Halogens) (+1, +3, +5 Oxidation number)



- Reverse of disproportionation is called Comproportionation. In some of the disproportionation reactions, by changing the medium (from acidic to basic or reverse), the reaction goes in backward direction and can be taken as an example of Comproportionation reaction.



Balancing of redox reactions

All balanced equations must satisfy two criteria.

1. Atom balance (mass balance) :

There should be the same number of atoms of each kind on reactant and product side.

2. Charge balance :

The sum of actual charges on both sides of the equation must be equal.

There are two methods for balancing the redox equations :

1. Oxidation – number change method
2. Ion electron method or half cell method

- Since First method is not very much fruitful for the balancing of redox reactions, students are advised to use second method (Ion electron method) to balance the redox reactions

Ion electron method :

By this method redox equations are balanced in two different medium.

- (a) Acidic medium (b) Basic medium

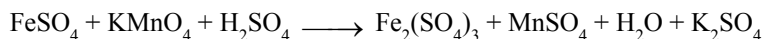
○ Balancing in acidic medium

Students are advised to follow the following steps to balance the redox reactions by ion electron method in acidic medium

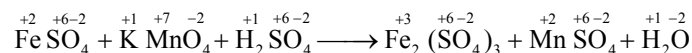
Solved Examples

Example - 3 :

Balance the following redox reaction :

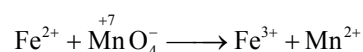


Sol. **Step-I** Assign the oxidation number to each element present in the reaction



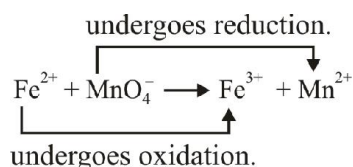
Step II :

Now convert the reaction in Ionic form by eliminating the elements or species, which are not undergoing either oxidation or reduction.



Step III :

Now identify the oxidation/reduction occurring in the reaction



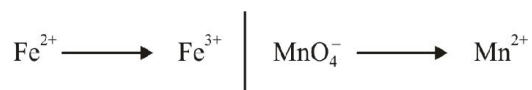
Step IV :

Split the Ionic reaction in two half, one for oxidation and other for reduction.



Step V :

Balance the atom other than oxygen and hydrogen atom in both half reactions



Fe & Mn atoms are balanced on both side.

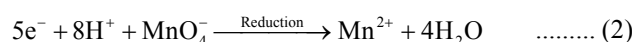
Step VI :

Now balance O & H atom by H_2O & H^+ respectively by the following way : For one excess oxygen atom, add one H_2O on the other side and two H^+ on the same side.



Step VII :

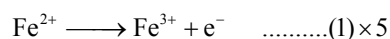
Equation (i) & (ii) are balanced atomwise. Now balance both equations chargewise. To balance the charge, add electrons to the electrically positive side.



Step VIII :

The number of electrons gained and lost in each half-reaction are equalised by multiplying both the half reactions with a suitable factor and finally the half reactions are added to give the overall balanced reaction.

Here, we multiply equation (1) by 5 and (2) by 1 and add them :

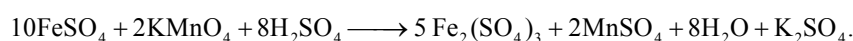
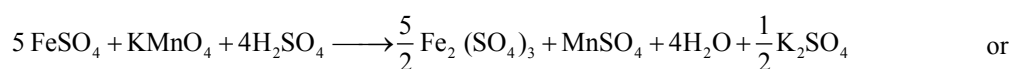


(Here, at this stage, you will get balanced redox reaction in Ionic form)

Step IX :

Now convert the ionic reaction into molecular form by adding the elements or species, which are removed in step (2).

Now, by some manipulation, you will get :



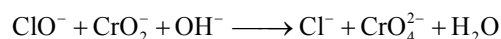
○ Balancing in basic medium :

In this case, except step VI, all the steps are same. We can understand it by the following example :

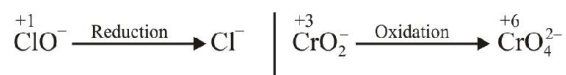
Solved Examples

Example - 4 :

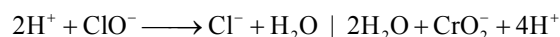
Balance the following redox reaction in basic medium :



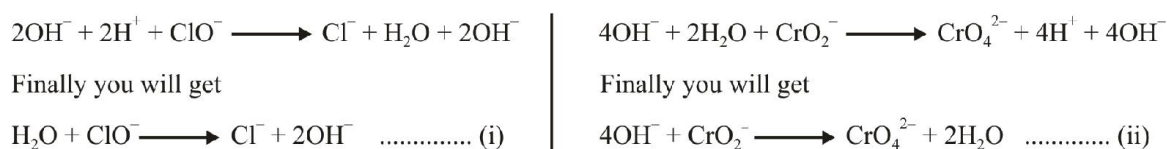
Sol. By using upto step V, we will get :



Now, students are advised to follow step VI to balance 'O' and 'H' atom.

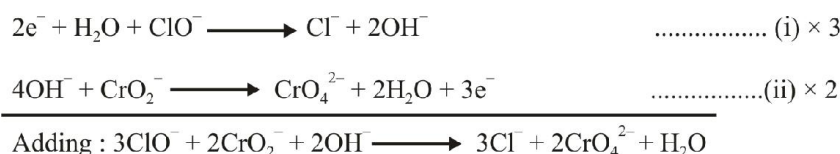


○ Now, since we are balancing in basic medium, therefore add as many as OH^- on both side of equation as there are H^+ ions in the equation.



Now see equation (i) and (ii) in which O and H atoms are balanced by OH^- and H_2O

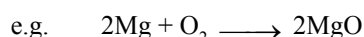
Now from step VIII



Concept of equivalents

Equivalent mass of element

Number of parts by mass of an element which reacts or displaces from a compound 1.008 parts by mass of hydrogen, 8 parts by mass of oxygen and 35.5 parts by mass of chlorine, is known as the equivalent weight of the element.



$$48\text{g} \quad 32\text{g}$$

$$12\text{g} \quad 8\text{g}$$

$\therefore$ 32g of O_2 reacts with 48 g of Mg

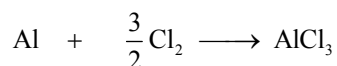
$$\therefore 8 \text{ g of } \text{O}_2 = \frac{48 \times 8}{32} = 12\text{g}$$

$\therefore$ Equivalent weight of Mg = 12



$$65.5 \text{ g} \quad 32.75$$

$$\therefore \text{Equivalent weight of Zn} = \frac{65.5}{2} = 32.75\text{g}$$



$$27\text{ g} \quad \frac{3}{2} \times 71\text{ g}$$

$\therefore$ 111.5 g chlorine reacts with 27 g of Al.

$$\therefore 35.5\text{ chlorine reacts with } \frac{27 \times 35.5}{111.5} = 9.0\text{ g of Al}$$

$$\therefore \text{Equivalent weight of aluminium} = \frac{27}{3} = 9.0$$

As we can see from the above examples that equivalent weight is the ratio of atomic weight and a factor (say n-factor or valency factor) which is in above three cases is their respective valencies.

Equivalent weight (E) :

$$\text{In general, Eq. wt. (E)} = \frac{\text{Atomic weight or molecular weight}}{\text{valency factor (v.f)}} = \frac{\text{Mol. wt.}}{\text{n - factor}} = \frac{M}{x}$$

$$\text{Number of Equivalents} = \frac{\text{mass of species}}{\text{eq. wt. of that species}}$$

For a solution, Number of equivalents = $N_1 V_1$, where N is the normality and V is the volume in litres

- Equivalent mass is a pure number which, when expressed in gram, is called gram equivalent mass.
- The equivalent mass of substance may have different values under different conditions.
- There is no hard and fast rule that equivalent weight will be always less than the molecular mass.

Valency factor calculation :

○ For Elements :

Valency factor = valency of the element.

○ For Acids :

Valency factor = number of replaceable H^+ ions per acid molecule

Solved Examples

Example - 5 : NaOH, KOH

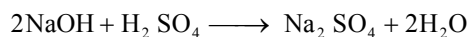
Sol. v.f. $\rightarrow$ 1 1

$$\text{Eq. wt.} \quad \frac{M}{1} \quad \frac{M}{1}$$

- Bases may be defined as the substances in which OH group is/are directly attached with group I elements (Li, Na, K, Rb, Cs), group II elements (Be, Mg, Ca, Ba) or group III elements (Al, Ga, In, Tl), transition metals, non-metallic cations like PH_4^+ , NH_4^+ etc.
- **Acid - base reaction :**
In case of acid base reaction, the valence factor is the actual number of H^+ or OH^- replaced in the reaction. The acid or base may contain more number of replaceable H^+ or OH^- than actually replaced in reaction.
- **v. f.** for base is the number of H^+ ion from the acid replaced by each molecule of the base

Solved Examples

Example - 6 :



Base Acid

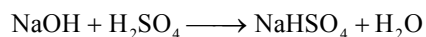
Sol. Valency factor of base = 1

Here, two molecule of NaOH replaced 2H^+ ion from the H_2SO_4 . Therefore, each molecule of NaOH replaced only one H^+ ion of acid, so v.f. = 1.

○ v. f. for acid is the number of OH^- replaced from the base by each molecule of acid

Solved Examples

Example - 7 :



Base Acid

Sol. Valency factor of acid = 1

Here, one of molecule of H_2SO_4 replaced one OH^- from NaOH. Therefore, valency factor for H_2SO_4 is one

$$\therefore \text{Eq. wt. of } \text{H}_2\text{SO}_4 = \frac{\text{Mol. wt}}{1}$$

○ **Salts :**

(a) **In non-reacting condition**

○ **Valency factor** = Total number of positive charge or negative charge present in the compound.

Solved Examples

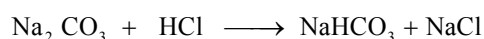
Example - 8 :

		Na_2CO_3 ,	$\text{Fe}_2(\text{SO}_4)_3$	$\text{FeSO}_4 \cdot 7\text{H}_2\text{O}$
Sol.	v.f.	2	$2 \times 3 = 6$	2
	Eq. wt.	$\frac{M}{2}$	$\frac{M}{6}$	$\frac{M}{2}$

(b) **In reacting condition**

Solved Examples

Example - 9 :



Base Acid

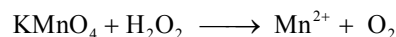
Sol. It is an acid base reaction, therefore valency factor for Na_2CO_3 is one while in non-reacting condition, it will be two.

(c) **Equivalent weight of oxidising / reducing agents in a redox reaction**

In case of redox change, v.f. = Total change in oxidation number per molecule.

Solved Examples

Example - 10 :



Sol. Mn in KMnO_4 is going from +7 to +2, so change in oxidation number per molecule of KMnO_4 is 5. So the valency factor of

KMnO_4 is 5 and equivalent weight is $\frac{M}{5}$.

Normality :

Normality of a solution is defined as the number of equivalents of solute present in one litre (1000 mL) solution.

Let V mL of a solution is prepared by dissolving W g of solute of equivalent weight E in water.

○ Number of equivalents of solute = $\frac{W}{E}$

V mL of solution contain $\frac{W}{E}$ equivalents of solute

∴ 1000 mL solution will contain $\frac{W \times 1000}{E \times V}$ equivalents of solute.

○ **Normality (N)** = $\frac{W \times 1000}{E \times V}$

○ **Normality (N) = Molarity × Valency factor**

$$N \times V \text{ (in mL)} = M \times V \text{ (in mL)} \times n$$

or

○ milliequivalents = millimoles × n

Solved Examples

Example 11 :

Calculate the normality of a solution containing 15.8 g of KMnO_4 in 50 mL acidic solution.

Sol : **Normality (N)** = $\frac{W \times 1000}{E \times V}$

Here $W = 15.8 \text{ g}$, $V = 50 \text{ mL}$ $E = \frac{\text{molar mass of } \text{KMnO}_4}{\text{Valency factor}} = 158/5 = 31.6$

So, normality = 10 N

Example 12 :

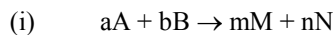
Calculate the normality of a solution containing 50 mL of 5 M solution of $\text{K}_2\text{Cr}_2\text{O}_7$ in acidic medium.

Sol : Normality (N) Molarity × valency factor = $5 \times 6 = 30 \text{ N}$

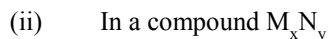
Law of Equivalence

The law states that one equivalent of an element combine with one equivalent of the other. In a chemical reaction, equivalents and milli equivalents of reactants react in equal amount to give same number of equivalents or milli equivalents of products separately.

Accordingly



$$\text{meq of A} = \text{meq of B} = \text{meq of M} = \text{m.eq. of N}$$

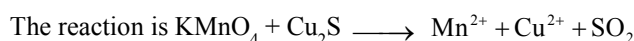


$$\text{meq of } M_xN_y = \text{meq of M} = \text{meq of N}$$

Solved Examples

Example 13 :

Find the number of moles of KMnO_4 needed to oxidise one mole Cu_2S in acidic medium.



Sol. From law of equivalence,

$$\text{equivalents of } \text{Cu}_2\text{S} = \text{equivalents of } \text{KMnO}_4$$

$$\text{moles of } \text{Cu}_2\text{S} \times \text{v.f.} = \text{moles of } \text{KMnO}_4 \times \text{v.f.}$$

$$1 \times 8 = \text{moles of } \text{KMnO}_4 \times 5 \quad \Rightarrow \text{moles of } \text{KMnO}_4 = 8/5$$

$$(\therefore \text{v.f. of } \text{Cu}_2\text{S} = 2(2-1) + 1(4 - (-2)) = 8 \text{ and v.f. of } \text{KMnO}_4 = 1(7-2) = 5)$$

Example 14 :

The number of moles of oxalate ions oxidized by one mole of MnO_4^- ion in acidic medium are :

(a) $\frac{5}{2}$

(b) $\frac{2}{5}$

(c) $\frac{3}{5}$

(d) $\frac{5}{3}$

Sol. Equivalents of $\text{C}_2\text{O}_4^{2-} = \text{equivalents of } \text{MnO}_4^-$

$$x(\text{mole}) \times 2 = 1 \times 5$$

$$(\therefore \text{v.f. of } \text{C}_2\text{O}_4^{2-} = 2(4-3) = 2 \text{ and v.f. of } \text{MnO}_4^- = 1(7-2) = 5.$$

$$x = \frac{5}{2} \text{ mole of } \text{C}_2\text{O}_4^{2-} \text{ ions.}$$

Solved Examples

Example : 15

How many millilitres of 0.02 M KMnO_4 solution would be required to exactly titrate 25 mL of 0.2 M $\text{Fe}(\text{NO}_3)_2$ solution in acidic medium ?

Sol. Method -1 : Mole concept method

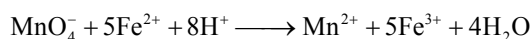
Starting with 25 mL of 0.2 M Fe^{2+} , we can write :

$$\text{Millimoles of } \text{Fe}^{2+} = 25 \times 0.2 \quad \dots\dots (1)$$

and in volume V (in milliliters) of the KMnO_4 ,

$$\text{Millimoles of } \text{MnO}_4^- = V(0.02) \quad \dots\dots\dots (2)$$

The balanced reaction is :



This requires that at the equivalent point,

$$\frac{\text{m.moles of MnO}_4^-}{1} = \frac{\text{m.moles of Fe}^{2+}}{5}$$

$$\therefore \frac{V(0.02)}{1} = \frac{(25)(0.2)}{5} \text{ (from (1) \& (2))}$$

$$\therefore V = 50 \text{ mL.}$$

Method -2 : Equivalent Method :

At the equivalence point,

$$\text{milliequivalents of MnO}_4^- = \text{milliequivalents of Fe}^{2+}$$

$$M_1 \times v f_1 \times V_1 = M_2 \times v f_2 \times V_2$$

$$0.02 \times 5 \times V_1 = 0.2 \times 1 \times 25 \quad (\because \text{MnO}_4^- \rightarrow \text{Mn}^{2+}; v.f. = 5, \text{Fe}^{2+} \rightarrow \text{Fe}^{3+}; v.f. = 1)$$

$$\therefore V_1 = 50 \text{ mL.}$$

Titration

Titration is procedure for determining the concentration of a solution by allowing a carefully measured volume to react with a standard solution of another substance, whose concentration is known.

Standard solution - It is a solution whose concentration is known and is taken in burette. It is also called **Titrant**.

There are two type of **titrants** :

- **Primary titrants/standard** - These reagents can be accurately weighed and their solutions are not to be standardised before use.

Ex : Oxalic acid, $\text{K}_2\text{Cr}_2\text{O}_7$, AgNO_3 , CuSO_4 , ferrous ammonium sulphate, hypo etc.

- **Secondarily titrants/standard** : These reagents cannot be accurately weighed and their solutions are to be standardised before use.

Ex : NaOH , KOH , HCl , H_2SO_4 , I_2 , KMnO_4 , etc.

Titrate : Solution consisting of substance to be estimated, generally taken in a beaker.

Equivalence point : It is the point when number of equivalents of titrant added becomes equal to number of equivalents of titrate.

At equivalence point :

$$n_1 V_1 M_1 = n_2 V_2 M_2$$

Indicator : An auxiliary substance added for physical detection of completion of titration at equivalence point. It generally show colour change on completion of titration.

Types of Titrations :

- Acid-base titrations (to be studied in Ionic equilibrium)

○ Redox Titrations

Some Common Redox Titrations

Table of Redox Titrations : (Excluding Iodometric/Iodimetric titrations)

	Estimation of	By titrating with	Reactions	Relation *between OA and RA
1.	Fe^{2+}	MnO_4^-	$\text{Fe}^{2+} \rightarrow \text{Fe}^{3+} + \text{e}^-$ $\text{MnO}_4^- + 8\text{H}^+ + 5\text{e}^- \rightarrow \text{Mn}^{2+} + 4\text{H}_2\text{O}$	$5\text{Fe}^{2+} \equiv \text{MnO}_4^-$ Eq. wt. of $\text{Fe}^{2+} = \text{M}/1$
2.	Fe^{2+}	$\text{Cr}_2\text{O}_7^{2-}$	$\text{Fe}^{2+} \rightarrow \text{Fe}^{3+} + \text{e}^-$ $\text{Cr}_2\text{O}_7^{2-} + 14\text{H}^+ + 6\text{e}^- \rightarrow 2\text{Cr}^{3+} + 7\text{H}_2\text{O}$	$6\text{Fe}^{2+} \equiv \text{Cr}_2\text{O}_7^{2-}$ Eq. wt. of $\text{Cr}_2\text{O}_7^{2-} = \text{M}/6$
3.	$\text{C}_2\text{O}_4^{2-}$	MnO_4^-	$\text{C}_2\text{O}_4^{2-} \rightarrow 2\text{CO}_2 + 2\text{e}^-$ $\text{MnO}_4^- + 8\text{H}^+ + 5\text{e}^- \rightarrow \text{Mn}^{2+} + 4\text{H}_2\text{O}$	$5\text{C}_2\text{O}_4^{2-} \equiv 2\text{MnO}_4^-$ Eq. wt. of $\text{C}_2\text{O}_4^{2-} = \text{M}/2$
4.	H_2O_2	MnO_4^-	$\text{H}_2\text{O}_2 \rightarrow 2\text{H}^+ + \text{O}_2 + 2\text{e}^-$ $\text{MnO}_4^- + 8\text{H}^+ + 5\text{e}^- \rightarrow \text{Mn}^{2+} + 4\text{H}_2\text{O}$	$5\text{H}_2\text{O}_2 \equiv 2\text{MnO}_4^-$ Eq. wt. of $\text{H}_2\text{O}_2 = \text{M}/2$
5.	As_2O_3	MnO_4^-	$\text{As}_2\text{O}_3 + 5\text{H}_2\text{O} \rightarrow 2\text{AsO}_4^{3-} + 10\text{H}^+ + 4\text{e}^-$ $\text{MnO}_4^- + 8\text{H}^+ + 5\text{e}^- \rightarrow \text{Mn}^{2+} + 4\text{H}_2\text{O}$	Eq. wt. of $\text{As}_2\text{O}_3 = \text{M}/4$
6.	AsO_3^{3-}	BrO_3^-	$\text{AsO}_3^{3-} + \text{H}_2\text{O} \rightarrow \text{AsO}_4^{3-} + 2\text{H}^+ + 2\text{e}^-$ $\text{BrO}_3^- + 6\text{H}^+ + 6\text{e}^- \rightarrow \text{Br}^- + 3\text{H}_2\text{O}$	Eq. wt. of $\text{AsO}_3^{3-} = \text{M}/2$ Eq. wt. of $\text{BrO}_3^- = \text{M}/6$

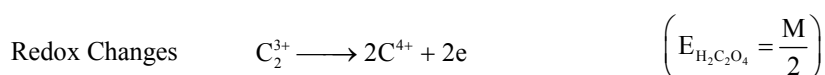
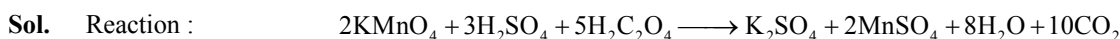
Permanganate Titrations :

- KMnO_4 is generally used as oxidising agent in acidic medium, generally provided by dilute H_2SO_4 .
- KMnO_4 works as self indicator persistent pink color is indication of end point.
- Mainly used for estimation of Fe^{2+} , oxalic acid, oxalates, H_2O_2 etc.

Solved Examples

Example 16 :

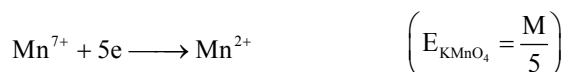
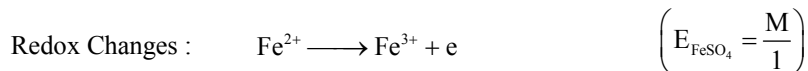
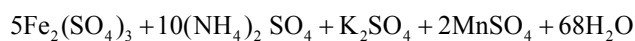
Write the balanced reaction of titration of KMnO_4 Vs oxalic acid in presence of H_2SO_4 .



Indicator : KMnO_4 acts as self indicator.

Example : 17

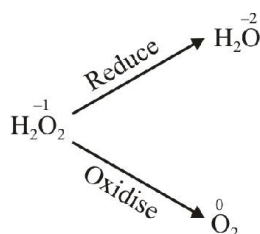
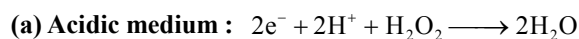
Write the balanced reaction of titration of KMnO_4 Vs ferrous ammonium sulphate in presence of H_2SO_4 .



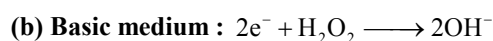
Indicator : KMnO_4 acts as self indicator.

Hydrogen peroxide (H_2O_2)

H_2O_2 can behave both like oxidising and reducing agent in both the mediums (acidic and basic).


○ Oxidising agent : ($\text{H}_2\text{O}_2 \rightarrow \text{H}_2\text{O}$)


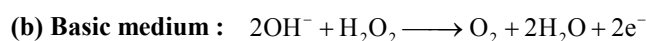
$$\text{v.f} = 2$$



$$\text{v.f} = 2$$

○ Reducing agent : ($\text{H}_2\text{O}_2 \rightarrow \text{O}_2$)


$$\text{v.f} = 2$$



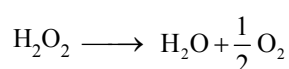
$$\text{v.f} = 2$$

Note : Valency factor of H_2O_2 is always equal to 2.

Volume strength of H_2O_2 : Strength of H_2O_2 is represented as 10V, 20V, 30V etc.

20V H_2O_2 means **one litre** of this sample of H_2O_2 on decomposition gives **20L of O_2** gas of STP.

Decomposition of H_2O_2 is given as :



$$1 \text{ mole} \quad \frac{1}{2} \times 22.4 \text{ L O}_2 \text{ at STP}$$

$$= 34\text{g} \quad = 11.2 \text{ L O}_2 \text{ at STP}$$

To obtain 11.2 litre O₂ at STP, at least 34 g H₂O₂ must be decomposed.

$$\text{For } 20 \text{ L O}_2, \text{ we should decompose atleast } \frac{34}{11.2} \times 20 \text{ g H}_2\text{O}_2$$

$$\therefore \quad 1\text{L solution of H}_2\text{O}_2 \text{ contains } \frac{34}{11.2} \times 20 \text{ g H}_2\text{O}_2$$

$$\therefore \quad 1\text{L solution of H}_2\text{O}_2 \text{ contains } \frac{34}{11.2} \times \frac{20}{17} \text{ equivalents of H}_2\text{O}_2 \quad (E_{\text{H}_2\text{O}_2} = \frac{M}{2} = \frac{34}{2} = 17)$$

$$\text{Normality of H}_2\text{O}_2 = \frac{34}{11.2} \times \frac{20}{17} = \frac{20}{5.6}$$

$$\bigcirc \quad \text{Normality of H}_2\text{O}_2 \text{ (N)} = \frac{\text{Volume strength of H}_2\text{O}_2}{5.6}$$

$$\therefore \quad M_{\text{H}_2\text{O}_2} = \frac{N_{\text{H}_2\text{O}_2}}{\text{v.f.}} = \frac{N_{\text{H}_2\text{O}_2}}{2}$$

$$\bigcirc \quad \text{Molarity of H}_2\text{O}_2 \text{ (M)} = \frac{\text{Volume strength of H}_2\text{O}_2}{11.2}$$

Strength (in g/L) : Denoted by S

$$\text{Strength} = \text{Molarity} \times \text{Mol. wt} = \text{Molarity} \times 34$$

$$\text{Strength} = \text{Normality} \times \text{Eq. weight} = \text{Normality} \times 17$$

Solved Examples

Example 18 :

20 mL of H₂O₂ after acidification with dilute H₂SO₄ required 30 mL of $\frac{N}{12}$ KMnO₄ for complete oxidation. Find the strength of H₂O₂ solution. [Molar mass of H₂O₂ = 34]

Sol. meq. of KMnO₄ = meq. of H₂O₂

$$30 \times \frac{1}{12} = 20 \times N'$$

$$N' = \frac{30}{12 \times 20} = \frac{1}{8} N$$

$$\therefore \quad \text{strength} = N' \times \text{equivalent mass} = \frac{1}{8} \times 17 = 2.12 \text{ g/L.}$$

Hardness of water (Hard water does not give lather with soap)

Temporary hardness - due to bicarbonates of Ca & Mg

Permanent hardness - due to chlorides & sulphates of Ca & Mg. There are some method by which we can soften the water sample.

- (a) By boiling : $2\text{HCO}_3^- \longrightarrow \text{H}_2\text{O} + \text{CO}_2 + \text{CO}_3^{2-}$ or
By Slaked lime : $\text{Ca}(\text{HCO}_3)_2 + \text{Ca}(\text{OH})_2 \longrightarrow \text{CaCO}_3 + 2\text{H}_2\text{O}$
 $\text{Ca}^{2+} + \text{CO}_3^{2-} \longrightarrow \text{CaCO}_3$
- (b) By Washing Soda : $\text{CaCl}_2 + \text{Na}_2\text{CO}_3 \longrightarrow \text{CaCO}_3 + 2\text{NaCl}$
- (c) By ion exchange resins : $\text{Na}_2\text{R} + \text{Ca}^{2+} \longrightarrow \text{CaR} + 2\text{Na}^+$
- (d) By adding chelating agents like $(\text{PO}_3^-)_3$ etc.

Parts Per Million (ppm)

When the solute is present in very less amount, then this concentration term is used. It is defined as the number of parts of the solute present in every 1 million parts of the solution. ppm can both be in terms of mass or in terms of moles. If nothing has been specified, we take ppm to be in terms of mass. Hence, a 100 ppm solution means that 100 g of solute is present in every 1000000g of solution.

$$\text{ppm}_A = \frac{\text{mass of A}}{\text{Total mass}} \times 10^6 = \text{mass fraction} \times 10^6$$

Measurement of Hardness :

Hardness is measured in terms of ppm (parts per million) of CaCO_3 or equivalent to it.

$$\text{Hardness in ppm} = \frac{\text{mass of CaCO}_3}{\text{Total mass of solution}} \times 10^6$$

Solved Examples

Example 19 :

0.00012% MgSO_4 and 0.000111% CaCl_2 is present in water. What is the measured hardness of water and millimoles of washing soda required to purify water 1000 L water ?

Sol. Basis of calculation = 100 g hard water

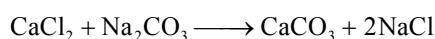
$$\text{MgSO}_4 = 0.00012\text{g} = \frac{0.00012}{120} \text{ mole}$$

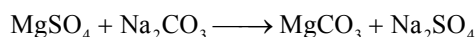
$$\text{CaCl}_2 = 0.000111\text{g} = \frac{0.000111}{111} \text{ mole}$$

$$\therefore \text{equivalent moles of CaCO}_3 = \left(\frac{0.00012}{120} + \frac{0.000111}{111} \right) \text{ mole}$$

$$\therefore \text{mass of CaCO}_3 = \left(\frac{0.00012}{120} + \frac{0.000111}{111} \right) \times 100 = 2 \times 10^{-4} \text{ g}$$

$$\text{Hardness (in terms of ppm of CaCO}_3) = \frac{2 \times 10^{-4}}{100} \times 10^6 = 2 \text{ ppm}$$





$$\therefore \text{Required Na}_2\text{CO}_3 \text{ for 100g of water} = \left(\frac{0.00012}{120} + \frac{0.000111}{111} \right) \text{mole}$$

$$= 2 \times 10^{-6} \text{ mole}$$

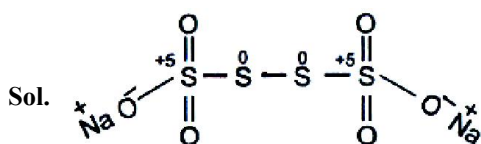
$$\therefore \text{Required Na}_2\text{CO}_3 \text{ for 1000 litre water} = \frac{2 \times 10^{-6}}{100} \times 10^6 = \frac{2}{100} \text{ mole} \quad (\because d = 1 \text{ g/mL})$$

$$= \frac{20}{1000} \text{ mole} = 20 \text{ m mole}$$

SOLVED EXAMPLES

Example – 1

Calculate individual oxidation number of each S-atom in $\text{Na}_2\text{S}_4\text{O}_6$ (sodium tetrathionate) with the help of its structure.



Example – 2

Find the average and individual oxidation number of Fe & Pb in Fe_3O_4 & Pb_3O_4 , which are mixed oxides.

Sol. (i) Fe_3O_4 is mixture of FeO & Fe_2O_3 in 1 : 1 ratio
so, individual oxidation number of Fe = +2 & +3

$$\text{& average oxidation number} = \frac{1(+2) + 2(+3)}{3} = 8/3$$

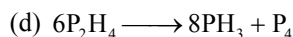
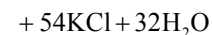
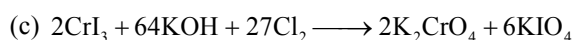
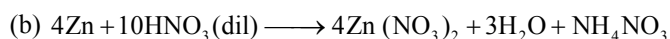
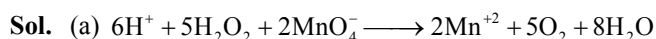
(ii) Pb_3O_4 is a mixture of PbO & PbO_2 in 2 : 1 molar ratio
so, individual oxidation number of Pb are +2 & +4

$$\text{& average oxidation number of Pb} = \frac{2(+2) + 1(+4)}{3} = 8/3$$

Example – 3

Balance the following equations :

- $\text{H}_2\text{O}_2 + \text{MnO}_4^- \longrightarrow \text{Mn}^{+2} + \text{O}_2$ (acidic medium)
- $\text{Zn} + \text{HNO}_3 \text{ (dil)} \longrightarrow \text{Zn}(\text{NO}_3)_2 + \text{H}_2\text{O} + \text{NH}_4\text{NO}_3$
- $\text{CrI}_3 + \text{KOH} + \text{Cl}_2 \longrightarrow \text{K}_2\text{CrO}_4 + \text{KIO}_4 + \text{KCl} + \text{H}_2\text{O}$
- $\text{P}_2\text{H}_4 \longrightarrow \text{PH}_3 + \text{P}_4$
- $\text{Ca}_3(\text{PO}_4)_2 + \text{SiO}_2 + \text{C} \longrightarrow \text{CaSiO}_3 + \text{P}_4 + \text{CO}$



Example – 4

Calculate the normality of a solution obtained by mixing 50 mL of 5M solution of $K_2Cr_2O_7$ and 50 mL of 2 M $K_2Cr_2O_7$ in acidic medium.

Sol. v.f. of $K_2Cr_2O_7 = 6$

$$\text{so } N_f = \frac{N_1 V_1 + N_2 V_2}{V_1 + V_2}$$

$$= \frac{5 \times 6 \times 50 + 2 \times 6 \times 50}{50 + 50} = 21 \text{ N}$$

Example – 5

Calculate the normality of a solution containing 13.4 g of sodium oxalate in 100 mL Sol.

Sol. Normality = $\frac{\text{wt. in g/eq. wt}}{\text{vol of solution in litre}}$

Here, eq. wt. of $Na_2C_2O_4 = 134/2 = 67$

$$\text{so } N = \frac{13.4/67}{100/1000} = 2 \text{ N}$$

Example – 6

The number of moles of ferrous oxalate oxidised by one mole of $KMnO_4$ in acidic medium is :

- (a) $\frac{5}{2}$ (b) $\frac{2}{5}$
(c) $\frac{3}{5}$ (d) $\frac{5}{3}$

Sol. Eq. of $FeC_2O_4 = \text{Eq. of } KMnO_4$
moles of $FeC_2O_4 \times 3 = \text{moles of } KMnO_4 \times 5$
so, moles of $FeC_2O_4 = 5/3$ **Ans. (d)**

Example – 7

How many moles of $KMnO_4$ are needed to oxidise a mixture of 1 mole of each $FeSO_4$ & FeC_2O_4 in acidic medium ?

- (a) $\frac{4}{5}$ (b) $\frac{5}{4}$
(c) $\frac{3}{4}$ (d) $\frac{5}{3}$

Sol. Eq. of $KMnO_4 = \text{Eq. of } FeSO_4 + \text{Eq. of } FeC_2O_4$
moles of $KMnO_4 \times 5 = \text{moles of } FeSO_4 \times 1 + \text{moles of } FeC_2O_4 \times 3$
 $\therefore$ moles of $KMnO_4 = 4/5$ **Ans. (a)**

Example – 8

A fresh H_2O_2 solution is labelled 11.2 V. This solution has the same concentration as a solution which is :

- (a) 3.4 % (w/w) (b) 3.4% (v/v)
(c) 3.4% (w/v) (d) None of these

Sol. Molarity of $H_2O_2 = \frac{\text{vol. strength}}{11.2} = \frac{11.2}{11.2} = 1$

$$\text{Now, \% (w/v)} = \frac{\text{wt. of solute in g}}{\text{wt. of solution in mL}} \times 100$$

$$= \text{Molarity} \times \text{Mol. wt. of solute} \times \frac{1}{10}$$

$$= 1 \times 34 \times \frac{1}{10} = 3.4\% \quad \text{Ans. (c)}$$

Example – 9

100 mL each of 1N H_2O_2 and 11.2 V H_2O_2 solution are mixed, then the final solution is equivalent to :

- (a) 3 M H_2O_2 solution (b) 0.5 N H_2O_2 solution
(c) 25.5 g/L H_2O_2 solution
(d) 2.55 g/L H_2O_2 Sol.

Sol. $N_{\text{final}} = \frac{N_1 V_1 + N_2 V_2}{V_1 + V_2} = \frac{1 \times 100 + \left(\frac{11.2}{5.6}\right) \times 100}{100 + 100} = 3/2 = 1.5 \text{ N}$

$$\text{So, Molarity} = \frac{\text{Normality}}{\text{v.f.}} = \frac{1.5}{2} = 0.75 \text{ M}$$

Strength of solution in g/L = Molarity $\times$ Mol. wt. = $0.75 \times 34 = 25.5 \text{ g/L}$ **Ans. (c)**

EXERCISE - 1 : BASIC OBJECTIVE QUESTIONS

Oxidizing/Reducing Agents

- The compound that can work both as an oxidising as well as a reducing agent is :
(a) KMnO_4 (b) H_2O_2
(c) $\text{Fe}_2(\text{SO}_4)_3$ (d) $\text{K}_2\text{Cr}_2\text{O}_7$
- Which of the following behaves as both oxidising and reducing agents?
(a) H_2SO_4 (b) SO_2
(c) H_2S (d) HNO_3
- Which of the following is not a redox reaction?
(a) $\text{CaCO}_3 \rightarrow \text{CaO} + \text{CO}_2$ (b) $\text{O}_2 + 2\text{H}_2 \rightarrow 2\text{H}_2\text{O}$
(c) $\text{Na} + \text{H}_2\text{O} \rightarrow \text{NaOH} + 1/2 \text{H}_2$
(d) $\text{MnCl}_3 \rightarrow \text{MnCl}_2 + 1/2 \text{Cl}_2$
- Which substance serves as reducing agent in the following reaction?
 $14\text{H}^+ + \text{Cr}_2\text{O}_7^{2-} + 3\text{Ni} \rightarrow 2\text{Cr}^{3+} + 7 \text{H}_2\text{O} + 3\text{Ni}^{2+}$
(a) H_2O (b) Ni
(c) H^+ (d) $\text{Cr}_2\text{O}_7^{2-}$
- Which of the following reactions depicts the oxidising property of SO_2 ?
(a) $\text{SO}_2 + \text{H}_2\text{O} \longrightarrow \text{H}_2\text{SO}_3$
(b) $2\text{H}_2\text{S} + \text{SO}_2 \longrightarrow 3\text{S} + 2\text{H}_2\text{O}$
(c) $\text{Cl}_2 + \text{SO}_2 \longrightarrow \text{SO}_2\text{Cl}_2$
(d) $2\text{MnO}_4^- + 5\text{SO}_2 + 2\text{H}_2\text{O} \longrightarrow 5\text{SO}_4^{2-} + 2\text{Mn}^{2+} + 4\text{H}^+$
- In which of the following reactions, there is no change in valency?
(a) $4\text{KClO}_3 \longrightarrow 3\text{KClO}_4 + \text{KCl}$
(b) $\text{SO}_2 + 2\text{H}_2\text{S} \longrightarrow 2\text{H}_2\text{O} + 3\text{S}$
(c) $\text{BaO}_2 + \text{H}_2\text{SO}_4 \longrightarrow \text{BaSO}_4 + \text{H}_2\text{O}_2$
(d) $2\text{BaO} + \text{O}_2 \longrightarrow 2\text{BaO}_2$
- Nitric oxide acts as a reducing agent in the reaction
(a) $4\text{NH}_3 + 5\text{O}_2 \longrightarrow 4\text{NO} + 6\text{H}_2\text{O}$
(b) $2\text{NO} + 3\text{I}_2 + 4\text{H}_2\text{O} \longrightarrow 2\text{NO}_3^- + 6\text{I}^- + 8\text{H}^+$
(c) $2\text{NO} + \text{H}_2\text{SO}_3 \longrightarrow \text{N}_2\text{O} + \text{H}_2\text{SO}_4$
(d) $2\text{NO} + \text{H}_2\text{S} \longrightarrow \text{N}_2\text{O} + \text{S} + \text{H}_2\text{O}$

Oxidation Number

- The oxidation state of osmium (Os) in OsO_4 is
(a) + 7 (b) + 5
(c) + 4 (d) + 8
- Oxidation number of nitrogen in $(\text{NH}_4)_2\text{SO}_4$ is
(a) $-1/3$ (b) -1
(c) + 1 (d) -3
- In which of the following compounds, the oxidation number of iodine is fractional?
(a) IF_7 (b) I_3^-
(c) IF_5 (d) IF_3
- The oxidation number of Phosphorus in $\text{Mg}_2\text{P}_2\text{O}_7$ is :
(a) + 3 (b) + 2
(c) + 5 (d) -3
- In which of the following compounds, nitrogen has an oxidation state of -1 ?
(a) N_2O (b) NO_2^-
(c) NH_2OH (d) N_2H_4
- In which of the following reactions is there a change in the oxidation number of nitrogen atom?
(a) $2 \text{NO}_2 \rightarrow \text{N}_2\text{O}_4$
(b) $\text{NH}_3 + \text{H}_2\text{O} \rightarrow \text{NH}_4^+ + \text{OH}^-$
(c) $\text{N}_2\text{O}_5 + \text{H}_2\text{O} \rightarrow 2\text{HNO}_3$
(d) None of these
- When SO_2 is passed through an acidified solution of potassium dichromate the oxidation state of S changes from:
(a) + 4 to 0 (b) + 4 to + 2
(c) + 4 to + 6 (d) + 6 to + 4
- Oxidation state of nitrogen is correctly given for

Compound	Oxidation state
(a) $[\text{Co}(\text{NH}_3)_5 \text{Cl}]\text{Cl}_2$	0
(b) NH_2OH	+ 1
(c) $(\text{N}_2\text{H}_5)_2\text{SO}_4$	+ 2
(d) Mg_3N_2	-3

16. The oxidation state of chromium in $\text{Cr}(\text{CO})_6$ is :
 (a) 0 (b) +2 (c) -2 (d) +6
17. In which of the following pairs, there is greatest difference in the oxidation number of the under lined elements?
 (a) $\underline{\text{N}}\text{O}_2$ and $\underline{\text{N}}_2\text{O}_4$ (b) $\underline{\text{P}}_2\text{O}_5$ and $\underline{\text{P}}_4\text{O}_{10}$
 (c) $\underline{\text{N}}_2\text{O}$ and $\underline{\text{N}}\text{O}$ (d) $\underline{\text{S}}\text{O}_2$ and $\underline{\text{S}}\text{O}_3$
18. In which of the compounds does manganese exhibit highest oxidation number?
 (a) MnO_2 (b) Mn_3O_4
 (c) K_2MnO_4 (d) MnSO_4
19. Which of the following elements has least oxidation number ?
 (a) $\text{Ni}(\text{CN})_4$ (b) $\text{Ni}(\text{CO})_4$
 (c) Fe_2O_3 (d) SF_6
20. The oxidation number of sulphur in S_8 , S_2F_2 , H_2S respectively are :
 (a) 0, +1 and -2 (b) +2, +1 and -2
 (c) 0, +1 and +2 (d) -2, +1 and -2
21. A metal ion M^{3+} loses 3 electrons, its oxidation number will be :
 (a) +3 (b) +6
 (c) 0 (d) -3
- (c) 5 16 2
 (d) 2 16 5
25. What is the coefficient of oxalate ion in the following reaction?
 $\text{MnO}_4^- + \text{C}_2\text{O}_4^{2-} + \text{H}^+ \longrightarrow \text{Mn}^{2+} + \text{CO}_2 + \text{H}_2\text{O}$
 (a) 4 (b) 2
 (c) 3 (d) 5
26. In the equation
 $\text{NO}_2^\ominus + \text{H}_2\text{O} \longrightarrow \text{NO}_3^\ominus + 2\text{H}^\oplus + \text{ne}^-$
 n stands for
 (a) 1 (b) 2
 (c) 3 (d) 4
27. In the chemical reaction,
 $\text{K}_2\text{Cr}_2\text{O}_7 + x\text{H}_2\text{SO}_4 + y\text{SO}_2 \longrightarrow \text{K}_2\text{SO}_4 + \text{Cr}_2(\text{SO}_4)_3 + z\text{H}_2\text{O}$
 x, y and z are
 (a) 1, 3, 1 (b) 4, 1, 4
 (c) 3, 2, 3 (d) 2, 1, 2

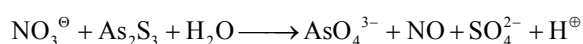
Equivalent Weight

Balancing of Redox Reactions

22. For the redox reaction $\text{MnO}_4^- + \text{C}_2\text{O}_4^{2-} + \text{H}^+ \rightarrow \text{Mn}^{2+} + \text{CO}_2 + \text{H}_2\text{O}$, The correct stoichiometric coefficients of MnO_4^- , $\text{C}_2\text{O}_4^{2-}$ and H^+ are respectively:
 (a) 2,5,16 (b) 16,5,2
 (c) 5,16,2 (d) 2,16,5
23. In the chemical reaction, $\text{K}_2\text{Cr}_2\text{O}_7 + x\text{H}_2\text{SO}_4 + y\text{SO}_2 \rightarrow \text{K}_2\text{SO}_4 + \text{Cr}_2(\text{SO}_4)_3 + z\text{H}_2\text{O}$, X, Y and Z are
 (a) 1,3,1 (b) 4,1,4
 (c) 3,2,3 (d) 2,1,2
24. For the redox reaction,
 $\text{MnO}_4^- + \text{C}_2\text{O}_4^{2-} + \text{H}^+ \longrightarrow \text{Mn}^{2+} + \text{CO}_2 + \text{H}_2\text{O}$ the correct coefficients of the reactants for the balanced reaction are:

MnO_4^-	$\text{C}_2\text{O}_4^{2-}$	H^+
(a) 2	5	16
(b) 16	5	2
28. In the conversion $\text{NH}_2\text{OH} \rightarrow \text{N}_2\text{O}$, the equivalent weight of NH_2OH will be :
 (a) $M/4$ (b) $M/2$
 (c) $M/5$ (d) $M/1$
 (M = molecular weight of NH_2OH)
29. The mass of oxalic acid crystals ($\text{H}_2\text{C}_2\text{O}_4 \cdot 2\text{H}_2\text{O}$) required to prepare 50 mL of a 0.2 N solution is :
 (a) 4.5 g (b) 6.3 g
 (c) 0.63 g (d) 0.45 g
30. M is molecular weight of KMnO_4 . The equivalent weight of KMnO_4 when it is converted into K_2MnO_4 is :
 (a) M (b) $M/3$
 (c) $M/5$ (d) $M/7$
31. The equivalent mass of oxidising agent in the following reaction is
 $\text{SO}_2 + 2\text{H}_2\text{S} \rightarrow 3\text{S} + 2\text{H}_2\text{O}$
 (a) 32 (b) 64
 (c) 16 (d) 8

32. In the following reaction :



the equivalent weight of As_2S_3 (with molecular weight M) is :

(a) $\frac{3M}{28}$ (b) $\frac{M}{4}$

(c) $\frac{M}{24}$ (d) $\frac{M}{28}$

33. Equivalent weight of H_3PO_2 (molecular weight = M) when it disproportionates into PH_3 and H_3PO_3 is

(a) M (b) $\frac{M}{2}$

(c) $\frac{M}{4}$ (d) $\frac{3M}{4}$

34. Equivalent weight of MnO_4^- in acidic, neutral and basic media are in ratio of :

(a) 3 : 5 : 15 (b) 5 : 3 : 1
(c) 5 : 1 : 13 (d) 3 : 15 : 5

35. K_2CrO_4 oxidises KI in the presence of HCl to I_2 . The equivalent weight of the K_2CrO_4 is

(a) $\frac{\text{Mw}}{2}$ (b) $\text{Mw} \times \frac{2}{3}$

(c) $\frac{\text{Mw}}{3}$ (d) $\frac{\text{Mw}}{6}$

36. The equivalent weight of MnSO_4 is half its molecular weight when it is converted to

(a) Mn_2O_3 (b) MnO_2
(c) MnO_4^- (d) MnO_4^{2-}

37. In the reaction $\text{VO} + \text{Fe}_2\text{O}_3 \rightarrow \text{FeO} + \text{V}_2\text{O}_5$ the eq. wt. of V_2O_5 is equal to its

(a) mol. wt. (b) mol. wt./8
(c) mol. wt./6 (d) none of these

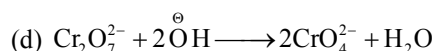
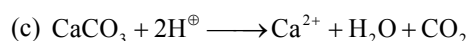
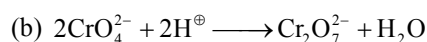
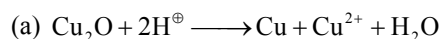
38. The eq. wt. of K_2CrO_4 as an oxidising agent in acid medium is

(a) mol. wt./2 (b) $\frac{2 \times \text{mol. wt.}}{3}$

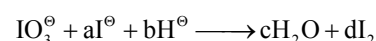
(c) $\frac{\text{mol. wt.}}{3}$ (d) $\frac{\text{mol. wt.}}{6}$

Disproportionation

39. Which of the following is a disproportionation reaction ?



40. In the balanced chemical reaction



a, b, c and d respectively, correspond to

(a) 5, 6, 3, 3 (b) 5, 3, 6, 3

(c) 3, 5, 3, 6 (d) 5, 6, 5, 5

Titration

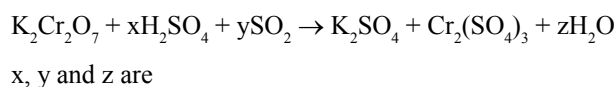
41. If equal volumes of 0.1 M KMnO_4 and 0.1 M $\text{K}_2\text{Cr}_2\text{O}_7$ solutions are allowed to oxidise Fe^{2+} to Fe^{3+} in acidic medium, then Fe^{2+} oxidised will be :

(a) More by KMnO_4 (b) More by $\text{K}_2\text{Cr}_2\text{O}_7$
(c) Equal in both cases (d) Cannot be determined.

42. Volume V_1 mL of 0.1M $\text{K}_2\text{Cr}_2\text{O}_7$ is needed for complete oxidation of 0.678 g N_2H_4 in acidic medium. The volume of 0.3 M KMnO_4 needed for same oxidation in acidic medium will be :

(a) $\frac{2}{5} V_1$ (b) $\frac{5}{2} V_1$
(c) $113 V_1$ (d) Can not be determined

43. In the chemical reaction,



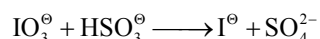
x, y and z are

(a) 1, 3, 1 (b) 4, 1, 4
(c) 3, 2, 3 (d) 2, 1, 2

44. 80 mL of KMnO_4 solution reacts with 3.4 g of $\text{Na}_2\text{C}_2\text{O}_4 \cdot 2\text{H}_2\text{O}$ in acidic medium. The molarity of the KMnO_4 solution is

(a) 0.5 M (b) 0.1 M
(c) 5 M (d) 1 M

45. What weight of NaHSO_3 is required to react with 100 mL of solution containing 0.33 g of NaIO_3 according to the following reaction



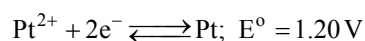
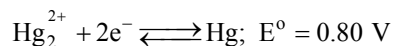
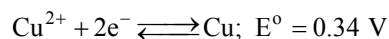
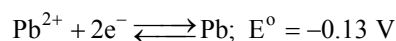
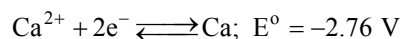
- (a) 0.52 g (b) 5.2 g
(c) 1.04 g (d) 10.4 g
46. How many moles of MnO_4^- ion will react with 1 mol of ferrous oxalate in acidic medium ?
- (a) $\frac{1}{5}$ (b) $\frac{2}{5}$
(c) $\frac{3}{5}$ (d) $\frac{5}{3}$
47. What volume of 0.05 M $\text{K}_2\text{Cr}_2\text{O}_7$ in acidic medium is needed for complete oxidation of 200 mL of 0.6 M FeC_2O_4 solution?
- (a) 1.2 mL (b) 1.2 L
(c) 120 mL (d) 800 mL
48. KI reacts with H_2SO_4 producing I_2 and H_2S . The volume of 0.2 M H_2SO_4 required to produce 0.1 mol of H_2S is
- (a) 4 L (b) 2.5 L
(c) 3.8 L (d) 5 L

Daniell Cell

49. Which of the following statement is correct for a galvanic cell ?
- (a) Reduction occurs at cathode
(b) Oxidation occurs at anode
(c) Electrons flow from anode to cathode
(d) All the statements are correct

Electrode Potential : Feasibility

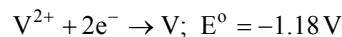
50. Consider the following standard reduction potentials:



Which of the following metals is the strongest REDUCING AGENT?

- (a) Ca (b) Pb
(c) Cu (d) Hg

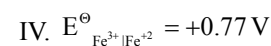
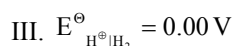
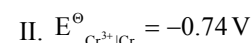
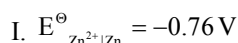
51. Consider the following electrodes potentials:



Which of the following reactions will proceed spontaneously from left to right?

- (a) $\text{Mg}^{2+} + \text{V} \rightarrow \text{Mg} + \text{V}^{2+}$
(b) $\text{Mg}^{2+} + 2\text{Cu}^+ \rightarrow \text{Mg} + 2\text{Cu}^{2+}$
(c) $\text{V}^{2+} + 2\text{Cu}^+ \rightarrow \text{V} + 2\text{Cu}^{2+}$
(d) $\text{V} + 2\text{Cu}^{2+} \rightarrow \text{V}^{2+} + 2\text{Cu}^+$

52. The standard reduction potentials at 298 K are given against each of the following half cell reactions :



Based on the above data, state which of the following is the strongest reducing agent ?

- (a) Zn (b) Cr
(c) H_2 (d) Fe

53. The correct order of reactivity of K, Mg, Zn and Cu with water according to the electrochemical series is

- (a) $\text{K} > \text{Mg} > \text{Zn} > \text{Cu}$ (b) $\text{Mg} > \text{Zn} > \text{Cu} > \text{K}$
(c) $\text{K} > \text{Zn} > \text{Mg} > \text{Cu}$ (d) $\text{Cu} > \text{Zn} > \text{Mg} > \text{K}$

H_2O_2

54. 34 g of H_2O_2 is present in 1120 mL of solution. This solution is called

- (a) 10 vol solution (b) 20 vol solution
(c) 34 vol solution (d) 32 vol solution

55. A 5.0 mL solution of H_2O_2 liberates 1.27 g of iodine from an acidified KI solution. The percentage strength of H_2O_2 is

- (a) 11.2 (b) 5.6
(c) 1.7 (d) 3.4

56. The volume strength of 1.5 N H_2O_2 solution is

- (a) 4.8 (b) 8.4
(c) 3.0 (d) 8.0

EXERCISE - 2 : PREVIOUS YEAR JEE MAINS QUESTIONS

- Which of the following reaction is possible at anode? (2002)
 - $F_2 + 2e^- \longrightarrow 2F^-$
 - $2H^+ + \frac{1}{2}O_2 + 2e^- \longrightarrow H_2O$
 - $2Cr_2^{3+} + 7H_2O \longrightarrow Cr_2O_7^{2-} + 14H^+ + 6e^-$
 - $Fe^{2+} \longrightarrow Fe^{3+} + e^-$
- Which of the following is a redox reaction (2002)
 - $NaCl + KNO_3 \longrightarrow NaNO_3 + KCl$
 - $CaC_2O_4 + 2HCl \longrightarrow CaCl_2 + H_2C_2O_4$
 - $Ca(OH)_2 + 2NH_4Cl \longrightarrow CaCl_2 + 2NH_3 + 2H_2O$
 - $2K[Ag(CN)_2] + Zn \longrightarrow 2Ag + K_2[Zn(CN)_4]$
- MnO_4^- is a good oxidising agent in different medium changing to

$$MnO_4^- \longrightarrow \begin{cases} Mn^{2+} \\ MnO_4^{2-} \\ MnO_2 \\ Mn_2O_3 \end{cases}$$

Changes in oxidation number respectively, are (2002)

 - 1, 3, 4, 5
 - 5, 4, 3, 2
 - 5, 1, 3, 4
 - 2, 6, 4, 3
- Oxidation number of Cl in $CaOCl_2$ (bleaching powder) is (2002)
 - zero, since it contains Cl_2
 - 1, since it contains Cl^-
 - +1, since it contains ClO^-
 - +1 and -1, since it contains ClO^- and Cl^-
- The oxidation state of chromium in the final product formed by the reaction between KI and acidified potassium dichromate solution is (2005)
 - +3
 - +2
 - +6
 - +4
- The oxidation state of Cr in $[Cr(NH_3)_4Cl_2]^+$ is (2005)
 - 0
 - +1
 - +2
 - +3
- Which of the following chemical reactions depicts the oxidising behaviour of H_2SO_4 ? (2006)
 - $2HI + H_2SO_4 \longrightarrow I_2 + SO_2 + 2H_2O$
 - $Ca(OH)_2 + H_2SO_4 \longrightarrow CaSO_4 + 2H_2O$
 - $NaCl + H_2SO_4 \longrightarrow NaHSO_4 + HCl$
 - $2PCl_5 + H_2SO_4 \longrightarrow 2POCl_3 + 2HCl + SO_2Cl_2$
- Consider the following reaction,

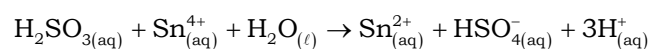
$$xMnO_4^- + yC_2O_4^{2-} + zH^+ \rightarrow xMn^{2+} + 2yCO_2 + \frac{z}{2}H_2O$$

The values of x, y and z in the reaction are, respectively (2013)

 - 5, 2 and 16
 - 2, 5 and 8
 - 2, 5 and 16
 - 5, 2 and 8
- Given, $E_{Cr^{3+}/Cr}^\circ = -0.74 V$; $E_{MnO_4^-/Mn^{2+}}^\circ = 1.51 V$
 $E_{Cr_2O_7^{2-}/Cr^{3+}}^\circ = 1.33 V$; $E_{Cl/Cl^-}^\circ = 1.36 V$
 Based on the data given above strongest oxidising agent will be (2013)
 - Cl
 - Cr^{3+}
 - Mn^{2+}
 - MnO_4^-
- Which of the following reactions is an example of a redox reaction? (2017)
 - $XeF_2 + PF_5 \rightarrow [XeF]^+ PF_6^-$
 - $XeF_6 + H_2O \rightarrow XeOF_4 + 2HF$
 - $XeF_6 + 2H_2O \rightarrow XeO_2F_2 + 4HF$
 - $XeF_4 + O_2F_2 \rightarrow XeF_6 + O_2$

JEE MAINS ONLINE QUESTION

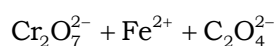
1. Consider the reaction :



Which of the following statements is correct?

Online 2014 SET (4)

- (a) H_2SO_3 is the reducing agent because it undergoes reduction
(b) Sn^{4+} is the reducing agent because it undergoes oxidation
(c) H_2SO_3 is the reducing agent because it undergoes oxidation
(d) Sn^{4+} is the oxidizing agent because it undergoes oxidation
2. How many electrons are involved in the following redox reaction ?



Online 2014 SET (4)

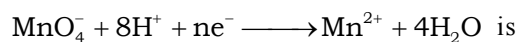
- (a) 3 (b) 4
(c) 5 (d) 6

3. What is the oxidation number of sulphur in $\text{Na}_2\text{S}_4\text{O}_6$?

Online 2016 SET (1)

- (a) 2/3 (b) 3/2
(c) 3/5 (d) 5/2

4. The value of n in



Online 2017 SET (1)

- (a) 5 (b) 4
(c) 2 (d) 3

5. In KO_2 , the nature of oxygen species and the oxidation state of oxygen atom are, respectively :

Online 2018 SET (2)

- (a) Oxide and -2
(b) Superoxide and $-\frac{1}{2}$
(c) Peroxide and
(d) Superoxide and -1

EXERCISE - 3 : ADVANCED OBJECTIVE QUESTIONS

1. All questions marked “S” are single choice questions
2. All questions marked “M” are multiple choice questions
3. All questions marked “C” are comprehension based questions
4. All questions marked “A” are assertion–reason type questions
 - (A) If both assertion and reason are correct and reason is the correct explanation of assertion.
 - (B) If both assertion and reason are true but reason is not the correct explanation of assertion.
 - (C) If assertion is true but reason is false.
 - (D) If reason is true but assertion is false.
5. All questions marked “X” are matrix–match type questions
6. All questions marked “I” are integer type questions

Oxidizing/Reducing Agents

1. (S) In the reaction, $2\text{FeCl}_3 + \text{H}_2\text{S} \longrightarrow 2\text{FeCl}_2 + 2\text{HCl} + \text{S}$
 - (a) FeCl_3 acts as an oxidizing agent
 - (b) Both H_2S and FeCl_3 are oxidized
 - (c) FeCl_3 is oxidised while H_2S is reduced
 - (d) H_2S acts as an oxidizing agent
2. (S) When KMnO_4 reacts with acidified FeSO_4
 - (a) Only FeSO_4 is oxidised
 - (b) Only KMnO_4 is oxidised
 - (c) FeSO_4 is oxidised and KMnO_4 is reduced
 - (d) None of the above
3. (S) Which of the following is a redox reaction?
 - (a) H_2SO_4 with NaOH
 - (b) In atmosphere, O_3 from O_2 by lightning
 - (c) Nitrogen oxides from nitrogen and oxygen by lightning
 - (d) Evaporation of H_2O
4. (M) Which of the following reactions do not involve oxidation or reduction:
 - (a) $2\text{Rb} + 2\text{H}_2\text{O} \longrightarrow 2\text{RbOH} + \text{H}_2$
 - (b) $2\text{CuI}_2 \longrightarrow 2\text{CuI} + \text{I}_2$
 - (c) $\text{NH}_4\text{Cl} + \text{NaOH} \longrightarrow \text{NaCl} + \text{NH}_3 + \text{H}_2\text{O}$
 - (d) $4\text{KCN} + \text{Fe}(\text{CN})_2 \longrightarrow \text{K}_4[\text{Fe}(\text{CN})_6]$

5. (M) Which of the following can act both as an oxidising as well as reducing agent :

- | | |
|--------------------------|----------------------------|
| (a) HNO_2 | (b) H_2O_2 |
| (c) H_2S | (d) SO_2 |

Oxidation Number

6. (A) **Assertion :** MnO_2 can act as an oxidizing agent as well as reducing agent.
Reason : Oxidation state of Mn lies between highest and lowest oxidation state.

(a) A	(b) B
(c) C	(d) D
7. (S) The oxidation number of cobalt in $\text{K}_3[\text{Co}(\text{NO}_2)_6]$ is

(a) 0	(b) + 4
(c) + 3	(d) + 6
8. (S) Phosphorus has the oxidation state of + 3 in

(a) Phosphorous acid	(b) Orthophosphoric
(c) Hypophosphorous acid	(d) Metaphosphoric acid
9. (I) It requires 40 mL of 1 M Ce^{4+} to titrate 20 mL of 1M Sn^{2+} to Sn^{4+} . What is the oxidation state of the Cerium in the product?
10. (M) When Cl_2 reacts with aqueous NaOH in cold condition then oxidation number of chlorine changes from 0 to:

(a) -1	(b) +1
(c) -2	(d) +2

Comprehension

The valency of carbon is generally 4, but its oxidation state may be -4, -2, 0, 2, -1 etc. In the compounds containing C, H and O, the oxidation number of C is calculated as

$$\text{Oxidation number of C} = \frac{2n_{\text{O}} - n_{\text{H}}}{n_{\text{C}}}$$

where n_{O} , n_{H} , n_{C} are the number of oxygen, hydrogen, and carbon atoms, respectively.

11. (C) In which of the following compounds is the valency of C two?

- (a) Ketenes (b) Alkenes
(c) Allenes (d) Carbenes

12. (C) In which of the following compounds is the oxidation state of C highest?

- (a) HCOOH (b) HCHO
(c) CH₃OH (d) CH₄

13. (C) In which of the following compounds is the oxidation state of C a fraction?

- (a) CO (b) CO₂
(c) Carbon suboxide (d) All

14. (A) **Assertion :** KO₂ is superoxide.

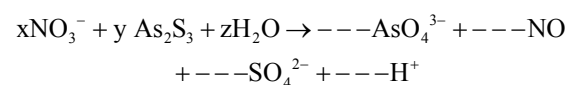
Reason : Oxidation state of oxygen of KO₂ is -2.

- (a) A (b) B
(c) C (d) D

15. (I) Oxidation number of chlorine in NOClO₄ is

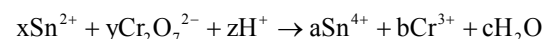
Balancing of Redox Reactions

16. (I) In the redox reaction,



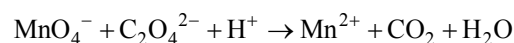
What is the value of $\frac{x}{z}$?

17. Dichromate ion in acidic medium oxidizes stannous ion as:



- (a) the value of x:y is 1 : 3 (b) the value of x+y+z is 18
(c) a:b is 3 : 2 (d) the value of z-c is 7

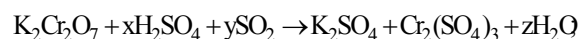
18. (S) For the redox reaction,



the correct coefficients of the reactions for the balanced reaction are respectively MnO_4^- , $\text{C}_2\text{O}_4^{2-}$, H^+ :

- (a) 2, 5, 16 (b) 16, 3, 12
(c) 15, 16, 12 (d) 2, 16, 5

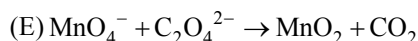
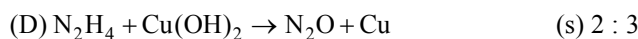
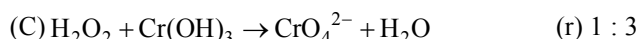
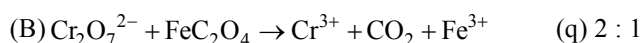
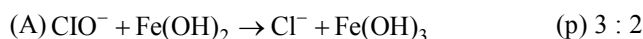
19. (S) In a chemical reaction



the value of x, y and z respectively are:

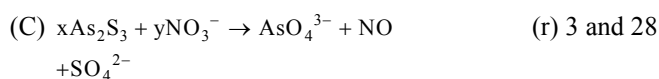
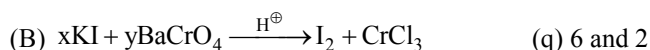
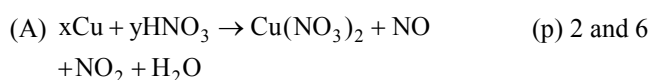
- (a) x = 1, y = 3, z = 1 (b) x = 4, y = 1, z = 4
(c) x = 3, y = 2, z = 1 (d) x = 2, y = 2, z = 1

20. (X) **Column I** **Column II**
(Redox reaction) (molar ratio of reducing to oxidising agent)



21. (X) Match the reaction in column I with the coefficients x and y given in column II.

- Column I** **Column II**
(Reaction) (The coefficients of x and y are)



22. (M) In the reaction
- $$I_2 + C_2H_5OH + \bar{O}H \rightarrow CHI_3 + HCOO^- + H_2O + I^-$$
- which of the following statements is/are correct?
- (a) The coefficients of $\bar{O}H$ and I^- in the given in balanced equation are, respectively, 6 and 5.
- (b) The coefficients of $\bar{O}H$ and I^- in the given balanced equation are, respectively, 5 and 6.
- (c) C_2H_5OH is oxidised to CHI_3 and $HCOO^-$
- (d) The number of electrons in the conversion of C_2H_5OH to CHI_3 and $HCOO^-$ is 8.

Equivalent Weight

23. (I) When BrO_3^- ion react with Br^- ion in acidic medium, Br_2 is liberated. Calculate the ratio of molecular weight and equivalent weight of $KBrO_3$.
24. (I) A volume of 12.5 mL of 0.05 M SeO_2 reacts with 25mL of 0.1M $CrSO_4$ which is oxidised to Cr^{3+} . To what oxidation state was the selenium converted by the reaction?

- | 25. (X) | Column I | Column II |
|--|---|-----------|
| (A) Eq. wt. = $\frac{\text{Molecular weight}}{33}$ | (p) When CrI_3 oxidises into $Cr_2O_7^{2-}$ and IO_4^- | |
| (B) Eq. wt. = $\frac{\text{Molecular weight}}{27}$ | (q) When $Fe(SCN)_2$ oxidises into Fe^{3+} , SO_4^{2-} , CO_3^{2-} , and NO_3^- | |
| (C) Eq. wt. = $\frac{\text{Molecular weight}}{28}$ | (r) When NH_4SCN oxidizes into SO_4^{2-} , CO_3^{2-} and NO_3^- | |
| (D) Eq. wt. = $\frac{\text{Molecular weight}}{24}$ | (s) When As_2S_3 oxidises into AsO_3^- and SO_4^{2-} | |

- | 26. (X) | Column I | Column II |
|--|--------------------------|-----------|
| (A) $P_2H_4 \rightarrow PH_3 + P_4H_2$ | (p) $E = \frac{3M}{4}$ | |
| (B) $I_2 \rightarrow I^- + IO_3^-$ | (q) $E = \frac{3M}{5}$ | |
| (C) $MnO_4^- + Mn^{2+} + H_2O \rightarrow \underline{Mn_3O_4} + H^+$ | (r) $E = \frac{15M}{26}$ | |
| (D) $H_3PO_2 \rightarrow PH_3 + H_3PO_3$ | (s) $E = \frac{5M}{6}$ | |

27. (M) When a equimolar mixture of Cu_2S and CuS is titrated with $Ba(MnO_4)_2$ in acidic medium, the final product's contains Cu^{2+} , SO_2 and Mn^{2+} . If the mol. wt. of Cu_2S , CuS and $Ba(MnO_4)_2$ are M_1 , M_2 and M_3 respectively then:

- (a) eq. wt. of Cu_2S is $\frac{M_1}{8}$
- (b) eq. wt. of CuS is $\frac{M_2}{6}$
- (c) eq. wt. of $Ba(MnO_4)_2$ is $\frac{M_3}{5}$
- (d) Cu_2S and CuS both have same equivalents in mixture

Comprehension

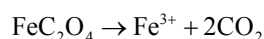
$$\text{Equivalent weight} = \frac{\text{Molecular weight / Atomic weight}}{n - \text{factor}}$$

n-factor is very important in redox as well as non-redox reactions. With the help of n-factor we can predicts the molar ratio of the reactant species taking part in reactions. The reciprocal of n-factor's ratio of the reactants is the molar ratio of the reactants.

In general n-factor of acid/base is number of moles of H^+/OH^- furnished per mole of acid/base. n-factor of a reactant is no. of moles of electrons lost or gained per mole of reactant.

Example 1 :

- In acidic medium : $KMnO_4 (n = 5) \rightarrow Mn^{2+}$
- In neutral medium : $KMnO_4 (n = 3) \rightarrow Mn^{2+}$
- In basic medium : $KMnO_4 (n = 1) \rightarrow Mn^{6+}$

Example 2 :


Total no. of moles of e^- lost by 1 mole of FeC_2O_4

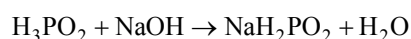
$$= 1 + 1 \times 2 \Rightarrow 3$$

$$\therefore \text{ n-factor of } \text{FeC}_2\text{O}_4 = 3$$

28. (C) n-factor of $\text{Ba}(\text{MnO}_4)_2$ in acidic medium is

- (a) 2 (b) 6
(c) 10 (d) None of these

29. (C) For the reaction,



What is the equivalent weight of H_3PO_2 ? (mol. wt. is M)

- (a) M (b) $M/2$
(c) $M/3$ (d) None of these

30. (C) For the reaction, $\text{Fe}_{0.95}\text{O}$ (molar mass: M) $\rightarrow \text{Fe}_2\text{O}_3$. What is the eq. wt. of $\text{Fe}_{0.95}\text{O}$?

- (a) $\frac{M}{0.85}$ (b) $\frac{M}{0.95}$
(c) $\frac{M}{0.8075}$ (d) None of these

31. (S) In the reaction between SO_2 and O_3 , the equivalent weight of ozone is :

- (a) The same as its molecular weight
(b) Half the molecular weight
(c) One-third of the molecular weight
(d) One-fourth of the molecular weight

32. (I) In the reaction : $\text{Na}_2\text{S}_2\text{O}_3 + 4\text{Cl}_2 + 5\text{H}_2\text{O} \longrightarrow \text{Na}_2\text{SO}_4 + \text{H}_2\text{SO}_4 + 8\text{HCl}$, the equivalent weight of $\text{Na}_2\text{S}_2\text{O}_3$ will be : (M = molecular weight of $\text{Na}_2\text{S}_2\text{O}_3$)

Disproportionation

33. (A) **Assertion :** F_2 does not undergo disproportionation reactions.

Reason : Fluorine shows only 0 and - 1 oxidation states.

- (a) A (b) B
(c) C (d) D

34. (M) Identify the correct statements with reference to the given reaction, $\text{P}_4 + 3\text{OH}^- + \text{H}_2\text{O} \rightarrow \text{PH}_3 + 3\text{H}_2\text{PO}_2^-$

- (a) Phosphorus is undergoing reduction only
(b) Phosphorus is undergoing oxidation only
(c) Phosphorus is undergoing oxidation as well as reduction.
(d) Hydrogen is undergoing neither oxidation nor reduction

35. (A) **Assertion :** White phosphorous reacts with aqueous caustic soda to form PH_3 and NaH_2PO_2 . It is disproportionation reaction.

Reason : In the reaction of disproportionation, same substance is oxidised as well as reduced simultaneously.

- (a) A (b) B
(c) C (d) D

Titrations

36. (S) 1 mole of equimolar mixture of ferric oxalate and ferrous oxalate will require x mole of KMnO_4 in acidic medium for complete oxidation, x is:

- (a) 0.5 mole (b) 0.9 mole
(c) 1.2 mole (d) 4.5 mole

37. (S) 20 mL of 0.2 M NaOH (aq) solution is mixed with 35 mL of 0.1 M NaOH (aq) solution and the resultant solution is diluted to 100 mL. 40 mL of this diluted solution reacted with 10% impure sample of oxalic acid ($\text{H}_2\text{C}_2\text{O}_4$). The weight of impure sample is:

- (a) 0.15 gram (b) 0.135 gram
(c) 0.59 gram (d) None of these

38. (S) A mixture of FeO and Fe_2O_3 is completely reacted with 100 mL of 0.25 M acidified KMnO_4 solution. The resultant solution was then titrated with Zn dust which converted Fe^{3+} of the solution of Fe^{2+} . The Fe^{2+} required 1000 mL of 0.10 M $\text{K}_2\text{Cr}_2\text{O}_7$ solution. Find out the weight % Fe_2O_3 in the mixture.

- (a) 80.85 (b) 19.15
(c) 50 (d) 89.41

39. (S) 2 mole, equimolar mixture of $\text{Na}_2\text{C}_2\text{O}_4$ and $\text{H}_2\text{C}_2\text{O}_4$ required V_1 L of 0.1 M KMnO_4 in acidic medium for complete oxidation. The same amount of the mixture required V_2 L of 0.2 M NaOH for neutralization. The ratio of V_1 to V_2 is:

- (a) 1 : 2 (b) 2 : 1
(c) 4 : 5 (d) 5 : 4

40. (S) 32 g of a sample of $\text{FeSO}_4 \cdot 7\text{H}_2\text{O}$ were dissolved in dilute sulphuric acid and water and its volume was made up to 1 litre, 25 mL of this solution required 20 mL of 0.02 M KMnO_4 solution for complete oxidation. Calculate the weight % of $\text{FeSO}_4 \cdot 7\text{H}_2\text{O}$ in the sample.
- (a) 34.75 (b) 69.5
(c) 89.5 (d) None of these
41. (S) 125 mL of 63% (w/v) $\text{H}_2\text{C}_2\text{O}_4 \cdot 2\text{H}_2\text{O}$ solution is made to react with 125 mL of a 40% (w/v) NaOH solution. The resulting solution is : (ignoring hydrolysis of ions)
- (a) Neutral (b) Acidic
(c) Strongly acidic (d) Alkaline
42. (S) An element A in a compound ABD has oxidation number $-n$. It is oxidised by $\text{Cr}_2\text{O}_7^{2-}$ in acidic medium. In the experiment, 1.68×10^{-3} moles of $\text{K}_2\text{Cr}_2\text{O}_7$ were used for 3.36×10^{-3} moles of ABD. The new oxidation number of A after oxidation is :
- (a) 3 (b) $3 - n$
(c) $n - 3$ (d) $+n$
43. (M) 25 mL of 0.5 M H_2O_2 solution is added to 50 mL of 0.2 M KMnO_4 in acid solution. Which of the following statements is false :
- (a) 0.010 mole of oxygen gas is liberated.
(b) 0.005 mole of KMnO_4 is left.
(c) 0.030 g of oxygen gas is evolved.
(d) 0.0025 mole H_2O_2 does not react with KMnO_4 .
44. (M) Which of the following solutions will exactly oxidize 25 mL of an acid solution of 0.1 M iron (II) oxalate :
- (a) 10 mL of 0.25 M KMnO_4
(b) 25 mL of 0.2 M KMnO_4
(c) 25 mL of 0.6 M KMnO_4
(d) 15 mL of 0.1 M KMnO_4
45. (M) In the following reaction : $\text{Cr}(\text{OH})_3 + \text{OH}^- + \text{IO}_3^- \rightarrow \text{CrO}_4^{2-} + \text{H}_2\text{O} + \text{I}^-$
- (a) IO_3^- is oxidising agent
(b) $\text{Cr}(\text{OH})_3$ is oxidised
(c) $6e^-$ are being taken per iodine atom
(d) None of these
46. (S) In the reaction, $8\text{Al} + 3\text{Fe}_3\text{O}_4 \rightarrow 4\text{Al}_2\text{O}_3 + 9\text{Fe}$, the number of electrons transferred from reductant to oxidant is:
- (a) 8 (b) 4
(c) 7 (d) 24
47. (S) In the reaction of sodium thiosulphate with I_2 in aqueous medium the equivalent weight of sodium thiosulphate is equal to:
- (a) molar mass of sodium thiosulphate
(b) the average molar masses of $\text{Na}_2\text{S}_2\text{O}_3$ and I_2
(c) half the molar masses of sodium thiosulphate
(d) twice of molar mass of sodium thiosulphate
48. (A) **Assertion :** If 1.50 mol of KMnO_4 is required for oxidation in acidic medium, 2.50 mol of KMnO_4 would be required for same oxidation but in basic medium.
- Reason :** In acidic medium, oxidation state of Mn changes from +7 to +2, while in basic medium, it changes from +7 to +4.
- (a) A (b) B
(c) C (d) D
49. (M) 500 mL of a 0.05 M Mohr salt solution required the same volume of permanganate solution for complete oxidation. Which of the followings is (are) true regarding the above redox reaction?
- (a) The molarity of KMnO_4 must be 0.01 M if the medium is acidic
(b) The molarity of KMnO_4 must be 0.0167 M if the medium is basic.
(c) The medium has no role to play in redox reaction.
(d) In the above reaction Fe(II) is oxidized to Fe(III) irrespective of medium.

Daniell Cell

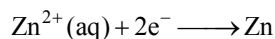
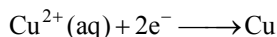
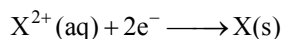
50. (M) In an electrochemical cell, a salt bridge is used:
- (a) to avoid mechanical contact of electrolyte solution in cathodic and anodic half-cell
(b) to maintain electrical neutrality in the cell
(c) to avoid liquid junction potential
(d) to mix the solution of anodic and cathodic half-cell

Electrode Potential : Feasibility

51. (S) Based on the following information arrange four metals A, B, C and D in order of decreasing ability to act as reducing agents:
- (I) Only A, B and C react with 1 M HCl to give H_2 (g)
(II) When C is added to solutions of the other metal ions, metallic B and D are formed
(III) Metal C does not reduce A^{n+} .
- (a) $\text{C} > \text{A} > \text{B} > \text{D}$ (b) $\text{C} > \text{A} > \text{D} > \text{B}$
(c) $\text{A} > \text{C} > \text{D} > \text{B}$ (d) $\text{A} > \text{C} > \text{B} > \text{D}$

Comprehension

The next two questions deal with an experiment. An unknown metal 'X' is found to react spontaneously with 1.0M solution of CuSO_4 plating out Cu(s) . X does not react with a solution of 1.0M $\text{Zn(NO}_3)_2$. The half-reactions for these metals are:



52. (C) What is the correct order for listing the metals according to decreasing strength as reducing agent ?

- (a) X, Cu, Zn (b) Cu, Zn, X
(c) Cu, X, Zn (d) Zn, X, Cu

53. (C) Another metal Y displaces Cu^{2+} from its aqueous solution but can't displace Zn^{2+} from its aqueous solution. Which of the following statements regarding X and Y is correct?

- (a) X is stronger reducing agent than Y
(b) X is weaker reducing agent than Y
(c) Both X and Y are weaker reducing agents than Cu
(d) X can be either stronger or weaker reducing agent than Y.

54. (S) If $\text{Sn}^{2+} + 2\text{e}^- \rightarrow \text{Sn}$; $E^\circ = -0.14 \text{ V}$



- (a) Sn^{2+} will disproportionate to Sn^{4+} and Sn
(b) Sn^{2+} is stable and it will not disproportionate
(c) Sn^{4+} is easily reduced to Sn
(d) none of the above

55. (M) Given that, $E^\circ_{\text{Ni}^{2+}/\text{Ni}} = -0.25 \text{ V}$, $E^\circ_{\text{Cu}^{2+}/\text{Cu}} = +0.34 \text{ V}$



Which of the following redox processes will not take place in specified direction?

- (a) $\text{Ni}^{2+}(\text{aq}) + \text{Cu(s)} \rightarrow \text{Ni(s)} + \text{Cu}^{2+}(\text{aq})$
(b) $\text{Cu(s)} + 2\text{Ag}^+(\text{aq}) \rightarrow \text{Cu}^{2+}(\text{aq}) + 2\text{Ag(s)}$
(c) $\text{Cu(s)} + 2\text{H}^+(\text{aq}) \rightarrow \text{Cu}^{2+}(\text{aq}) + \text{H}_2(\text{g})$
(d) $\text{Zn(s)} + 2\text{H}^+(\text{aq}) \rightarrow \text{Zn}^{2+}(\text{aq}) + \text{H}_2(\text{g})$

H_2O_2

56. (I) 20 mL of H_2O_2 after acidification with dilute H_2SO_4 required 30 mL of $\text{N}/12 \text{ KMnO}_4$ for complete oxidation. The strength of H_2O_2 solution is in g/L : [Molar mass of $\text{H}_2\text{O}_2 = 34$]

57. (I) A sample of 28 mL of $\text{H}_2\text{O}_2(\text{aq})$ solution required 10 mL of 0.1 $\text{KMnO}_4(\text{aq})$ solution for complete reaction in acidic medium. What is the volume strength of H_2O_2 ?

58. (S) A bottle of H_2O_2 is labelled as 10 vol H_2O_2 . 112 mL of this solution of H_2O_2 is titrated against 0.04 M acidified solution of KMnO_4 . The volume of KMnO_4 in litre is

- (a) 1 L (b) 2 L
(c) 3 L (d) 4 L

EXERCISE - 4 : PREVIOUS YEAR JEE ADVANCED QUESTION

- The equivalent weight of MnSO_4 is half of its molecular weight when it converts to (1988)
(a) Mn_2O_3 (b) MnO_2
(c) MnO_4^- (d) MnO_4^{2-}
- The oxidation number of phosphorus in $\text{Ba}(\text{H}_2\text{PO}_2)_2$ is (1988)
(a) +3 (b) +2
(c) +1 (d) -1
- The volume strength of 1.5 N H_2O_2 is (1990)
(a) 4.8 (b) 8.4
(c) 3.0 (d) 8.0
- For the redox reaction
 $\text{MnO}_4^- + \text{C}_2\text{O}_4^{2-} + \text{H}^+ \rightarrow \text{Mn}^{2+} + \text{CO}_2 + \text{H}_2\text{O}$
The correct coefficients of the reactants for the balanced reaction are

MnO_4^-	$\text{C}_2\text{O}_4^{2-}$	H^+	(1992)
(a) 2	5	16	
(b) 16	5	2	
(c) 5	16	2	
(d) 2	16	5	
- The number of moles of KMnO_4 that will be needed to react with one mole of sulphite ion in acidic solution (1997)
(a) $\frac{2}{5}$ (b) $\frac{3}{5}$
(c) $\frac{4}{5}$ (d) 1
- The number of moles of KMnO_4 that will be needed to react completely with one mole of ferrous oxalate in acidic medium is (1997)
(a) $\frac{2}{5}$ (b) $\frac{3}{5}$
(c) $\frac{4}{5}$ (d) 1
- The normality of 0.3 M phosphorus acid (H_3PO_3) is (1999)
(a) 0.1 (b) 0.9
(c) 0.3 (d) 0.6
- The oxidation number of sulphur in S_8 , S_2F_2 , H_2S respectively, are (1999)
(a) 0, +1 and -2 (b) +2, +1 and -2
(c) 0, +1 and +2 (d) -2, +1 and -2
- Among the following, the species in which the oxidation number of an element is +6 (2000)
(a) MnO_4^- (b) $\text{Cr}(\text{CN})_6^{3-}$
(c) NiF_6^{2-} (d) CrO_2Cl_2
- An aqueous solution of 6.3 g oxalic acid dihydrate is made up to 250 mL. The volume of 0.1 N NaOH required to completely neutralize 10 mL of this solution is (2000)
(a) 40 mL (b) 20 mL
(c) 10 mL (d) 4 mL
- The reaction, $3\text{ClO}(\text{aq})^- \longrightarrow \text{ClO}_3(\text{aq})^- + 2\text{Cl}(\text{aq})^-$ is an example of (2001)
(a) oxidation reaction
(b) reduction reaction
(c) disproportionation reaction
(d) decomposition reaction
- In the standardization of $\text{Na}_2\text{S}_2\text{O}_3$ and $\text{K}_2\text{Cr}_2\text{O}_7$ by iodometry, the equivalent weight of $\text{K}_2\text{Cr}_2\text{O}_7$ is (2000)
(a) (molecular weight)/2 (b) (molecular weight)/6
(c) (molecular weight)/3 (d) same as molecular weight
- Consider a titration of potassium dichromate solution with acidified Mohr's salt solution using diphenylamine as indicator. The number of moles of Mohr's salt required per mole of dichromate is (2007)
(a) 3 (b) 4
(c) 5 (d) 6

14. Which ordering of compounds is according to the decreasing order of the oxidation state of nitrogen ?

(2012)

- (a) $\text{HNO}_3, \text{NO}, \text{NH}_4\text{Cl}, \text{N}_2$
(b) $\text{HNO}_3, \text{NO}, \text{N}_2, \text{NH}_4\text{Cl}$
(c) $\text{HNO}_3, \text{NH}_4\text{Cl}, \text{NO}, \text{N}_2$
(d) $\text{NO}, \text{HNO}_3, \text{NH}_4\text{Cl}, \text{N}_2$

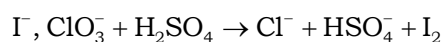
15. The reaction of white phosphorus with aqueous NaOH gives phosphine along with another phosphorus containing compound. The reaction type; the oxidation states of phosphorus in phosphine and the other product are respectively

(2013)

- (a) redox reaction; -3 and -5
(b) redox reaction; 3 and $+5$
(c) disproportionation reaction; -3 and $+5$
(d) disproportionation reaction; -3 and $+3$

16. For the reaction :

(2016)



The correct statement(s) in the balanced equation is/are:

- (a) Stoichiometric coefficient of HSO_4^- is 6.
(b) Iodide is oxidized
(c) Sulphur is reduced
(d) H_2O is one of the products.

Assertion and Reason

17. **Assertion :** In the titration of Na_2CO_3 with HCl using methyl orange indicator, the volume required at the equivalence point is twice that of the acid required using phenolphthalein indicator.

Reason : Two moles of HCl are required for the complete neutralization of one mole of Na_2CO_3 . (1991)

- (a) Assertion is true; Reason is true; Reason is the correct explanation of Assertion.
(b) Assertion is true; Reason is true; Reason is not the correct explanation of Assertion.
(c) Assertion is true; Reason is false.
(d) Assertion is false; Reason is true.

Fill in the Blanks

18. The compound $\text{YBa}_2\text{Cu}_3\text{O}_7$, which shows super conductivity, has copper in oxidation state..... Assume that the rare earth element yttrium is in its usual $+3$ oxidation state. (1994)

Integer Type Questions

19. A student performs a titration with different burettes and finds titre values of 25.2 mL, 25.25 mL, and 25.0 mL. The number of significant figures in the average titre value is (2010)

20. Among the following, the number of elements showing only one non-zero oxidation state is O, Cl, F, N, P, Sn, Tl, Na, Ti (2010)

21. The difference in the oxidation numbers of the two types of sulphur atoms in $\text{Na}_2\text{S}_4\text{O}_6$ is (2011)

Subjective Questions

22. The density of a 3 M sodium thiosulphate solution ($\text{Na}_2\text{S}_2\text{O}_3$) is 1.25 g per mL. Calculate (i) the percentage by weight of sodium thiosulphate (ii) the mole fraction of sodium thiosulphate and (iii) the molalities of Na^+ and $\text{S}_2\text{O}_3^{2-}$ ions. (1983)

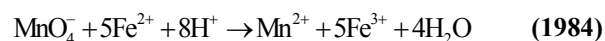
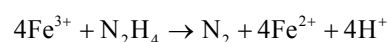
23. 4.08 g of a mixture of BaO and unknown carbonate MCO_3 was heated strongly. The residue weighed 3.64 g. This was dissolved in 100 mL of 1 N HCl. The excess acid required 16 mL of 2.5 N NaOH solution for complete neutralization. Identify the metal M. (1983)

24. 2.68×10^{-3} moles of a solution containing an ion A^{n+} require 1.61×10^{-3} moles of MnO_4^- for the oxidation of A^{n+} to AO_3^- in acidic medium. What is the value of n? (1984)

25. 5 mL of 8 N nitric acid, 4.8 mL of 5 N hydrochloric acid and a certain volume of 17 M sulphuric acid are mixed together and made up to 2 L. 30 mL of this acid mixture exactly neutralise 42.9 mL of sodium carbonate solution containing one gram of $\text{Na}_2\text{CO}_3 \cdot 10\text{H}_2\text{O}$ in 100 mL of water. Calculate the amount in gram of the sulphate ions in solution. (1985)

26. A sample of hydrazine sulphate ($\text{N}_2\text{H}_6\text{SO}_4$) was dissolved in 100 mL of water, 10 mL of this solution was reacted with excess of ferric chloride solution and warmed to complete the reaction. Ferrous ion formed was estimated and it, required 20 mL of M/50 potassium permanganate solution. Estimate the amount of hydrazine sulphate in one litre of the solution

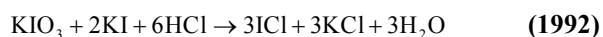
Reaction:



27. An equal volume of a reducing agent is titrated separately with 1 M KMnO_4 in acid, neutral and alkaline medium. The volumes of KMnO_4 required are 20 mL in acid, 33.3 mL in neutral and 100 mL in alkaline media. Find out the oxidation state of manganese in each reduction product. Give the balanced equations for all the three half reaction. Find out the volume of 1 M $\text{K}_2\text{Cr}_2\text{O}_7$ consumed, if the same volume of the reducing agent is titrated in acid medium. (1989)

28. An organic compound X on analysis gives 24.24 per cent carbon and 4.04 per cent hydrogen. Further, sodium extract of 1.0 g of X gives 2.90 g of silver chloride with acidified silver nitrate solution. The compound X may be represented by two isomeric structures Y and Z. Y on treatment with aqueous potassium hydroxide solution gives a dihydroxy compound while Z on similar treatment gives ethanal. Find out the molecular formula of X and gives the structure of Y and Z. (1989)
29. A mixture of $\text{H}_2\text{C}_2\text{O}_4$ (oxalic acid) and NaHC_2O_4 weighing 2.02 g was dissolved in water and the solution made up to one litre. Ten millilitres of the solution required 3.0 mL of 0.1 N sodium hydroxide solution for complete neutralization. In another experiment, 10.0 mL of the same solution, in hot dilute sulphuric acid medium, required 4.0 mL of 0.1 N potassium permanganate solution for complete reaction. Calculate the amount of $\text{H}_2\text{C}_2\text{O}_4$ and NaHC_2O_4 in the mixture. (1990)
30. A solution of 0.2 g of a compound containing Cu^{2+} and $\text{C}_2\text{O}_4^{2-}$ ions on titration with 0.02 M KMnO_4 in presence of H_2SO_4 consumes 22.6 mL of the oxidant. The resultant solution is neutralized with Na_2CO_3 , acidified with dilute acetic acid and treated with excess KI. The liberated iodine requires 11.3 mL of 0.05 M $\text{Na}_2\text{S}_2\text{O}_3$ solution for complete reduction. Find out the mole ratio of Cu^{2+} to $\text{C}_2\text{O}_4^{2-}$ in the compound. Write down the balanced redox reactions involved in the above titrations. (1991)
31. A 1.0 g sample of Fe_2O_3 solid of 55.2% purity is dissolved in acid and reduced by heating the solution with zinc dust. The resultant solution is cooled and made up to 100.0 mL. An aliquot of 25.0 mL of this solution requires for titration. Calculate the number of electrons taken up by the oxidant in the reaction of the above titration. (1991)
32. A 2.0 g sample of a mixture containing sodium carbonate, sodium bicarbonate and sodium sulphate is gently heated till the evolution of CO_2 ceases. The volume of CO_2 at 750 mm Hg pressure and at 298 K is measured to be 123.9 mL. A 1.5 g of the same sample requires 150 mL of (M/10) HCl for complete neutralization. Calculate the percentage composition of the components of the mixture. (1992)
33. One gram of commercial AgNO_3 is dissolved in 50 mL of water. It is treated with 50 mL of a KI solution. The silver iodide thus precipitated is filtered off. Excess of KI in the filtrate is titrated with (M/10) KIO_3 solution in presence of 6 M HCl till all I^- ions are converted into ICl . It requires 50 mL of (M/10) KIO_3 solution. 20 mL of the same stock solution of KI requires 30 mL of (M/10) KIO_3 under similar conditions. Calculate the percentage of AgNO_3 in the sample.
34. A 5.0 cm^3 solution of H_2O_2 liberates 0.508 g of iodine from an acidified KI solution. Calculate the strength of H_2O_2 solution in terms of volume strength at STP (1995)
35. A 20.0 cm^3 mixture of CO, CH_4 and He gases is exploded by an electric discharge at room temperature with excess of oxygen. The volume contraction is found to be 13.0 cm^3 . A further contraction of 14.0 cm^3 occurs when the residual gas is treated with KOH solution. Find out the composition of the gaseous mixture in terms of volume percentage. (1995)
36. A 3.00 g sample containing Fe_3O_4 , Fe_2O_3 and an inert impure substance, is treated with excess of KI solution in presence of dilute H_2SO_4 . The entire iron is converted into Fe^{2+} along with liberation of iodine. The resulting solution is diluted to 100 mL. A 20 mL of the dilute solution requires 11.0 mL of 0.5 M $\text{Na}_2\text{S}_2\text{O}_3$ solution to reduce the iodine present. A 50 mL of the dilute solution, after complete extraction of the iodine required 12.80 mL of 0.25 M KMnO_4 solution in dilute H_2SO_4 medium for the oxidation of Fe^{2+} . Calculate the percentage of Fe_2O_3 and Fe_3O_4 in the original sample. (1996)
37. To a 25 mL H_2O_2 solution, excess of acidified solution of potassium iodide was added. The iodine liberated required 20 mL of 0.3 N sodium thiosulphate solution. Calculate the volume strength of H_2O_2 solution. (1997)
38. An aqueous solution containing 0.10 g KIO_3 (formula weight = 214.0) was treated with an excess of KI solution. The solution was acidified with HCl. The liberated I_2 consumed 45.0 mL of thiosulphate solution decolourise the blue starch-iodine complex. Calculate the molarity of the sodium thiosulphate solution. (1998)
39. How many millilitres of 0.5 M H_2SO_4 are needed to dissolve 0.5 g of copper (II) carbonate? (1999)
40. Hydrogen peroxide solution (20 mL) reacts quantitatively with a solution of KMnO_4 (20 mL) acidified with dilute H_2SO_4 . The same volume of the KMnO_4 solution is just decolourised by 10 mL of MnSO_4 in neutral medium simultaneously forming a dark brown precipitate of hydrated MnO_2 . The brown precipitate is dissolved in 10 mL of 0.2 M sodium oxalate under boiling condition in the presence of dilute H_2SO_4 . Write the balanced equation involved in the reactions and calculate the molarity of H_2O_2 (2001)
41. Calculate the amount of calcium oxide required when it reacts with 852 g of P_4O_{10} . (2005)

Reaction :



ANSWER KEY

Exercise - 1 : (Basic Objective Questions)

1. (b)	2. (b)	3. (a)	4. (b)	5. (b)	6. (c)	7. (b)	8. (d)	9. (d)	10. (b)
11. (c)	12. (c)	13. (d)	14. (c)	15. (d)	16. (a)	17. (d)	18. (c)	19. (b)	20. (a)
21. (b)	22. (a)	23. (a)	24. (a)	25. (d)	26. (b)	27. (a)	28. (b)	29. (c)	30. (a)
31. (c)	32. (d)	33. (d)	34. (a)	35. (c)	36. (b)	37. (c)	38. (c)	39. (a)	40. (a)
41. (b)	42. (a)	43. (a)	44. (b)	45. (a)	46. (c)	47. (b)	48. (b)	49. (d)	50. (a)
51. (d)	52. (a)	53. (a)	54. (a)	55. (d)	56. (b)				

Exercise - 2 : (Previous Year JEE Mains Questions)

1. (cd)	2. (d)	3. (c)	4. (d)	5. (a)	6. (d)	7. (a)	8. (c)	9. (d)	10. (d)
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JEE Mains Online

1. (c)	2. (d)	3. (d)	4. (a)	5. (b)
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Exercise - 3 : (Advanced Objective Questions)

1. (a)	2. (c)	3. (c)	4. (cd)	5. (abd)	6. (a)	7. (c)	8. (a)	9. + 3	10. (ab)
11. (d)	12. (a)	13. (c)	14. (c)	15. (0007)	16. (0007)	17. (bcd)	18. (a)	19. (a)	
20. $A \rightarrow q; B \rightarrow q; C \rightarrow s; D \rightarrow r; E \rightarrow p$	21. $A \rightarrow p; B \rightarrow q; C \rightarrow r; D \rightarrow s; E \rightarrow t$	22. (acd)	23. (0005)						
24. (0000)	25. $A \rightarrow q; B \rightarrow p; C \rightarrow s; D \rightarrow r$	26. $A \rightarrow s; B \rightarrow q; C \rightarrow r; D \rightarrow p$	27. (ab)	28. (c)					
29. (a)	30. (a)	31. (b)	32. M/8	33. (a)	34. (cd)	35. (a)	36. (b)	37. (a)	38. (a)
39. (c)	40. (b)	41. (a)	42. (b)	43. (acd)	44. (d)	45. (abc)	46. (d)	47. (a)	48. (a)
49. (abd)	50. (abc)	51. (d)	52. (d)	53. (d)	54. (b)	55. (ac)	56. 2 g/L	57. (0001)	
58. (a)									

Exercise - 4 : (Previous Year JEE Advanced Questions)

1. (b)	2. (c)	3. (b)	4. (a)	5. (a)	6. (b)	7. (d)	8. (a)	9. (d)	10. (a)
11. (c)	12. (b)	13. (d)	14. (b)	15. (c)	16. (b)	17. (a,b,d)	18. $\frac{7}{3}$	19. (0003)	20. (0002)
21. (0005)	22. (i) 37.92, (ii) 0.065, (iii) 7.73 and 3.86	23. Calcium	24. 2	25. 6.5g	26. 6.5g/L	27. 16.67 mL			
28. Leave	29. 1.12g	30. 1 : 2	31. 1.04×10^{21}	32. 26.5%	33. 85%	34. 4.48%	35. 50%, 20%, 30%		
36. 49.33%, 34.8%	37. 1.33V	38. 0.06	39. 8.1 mL	40. 0.1 M	41. 10.08g				

Dream on !!

